

Trade-off analysis for optimal design of trigeneration energy systems integrated with solar and hydrogen subsystems: A novel optimization strategy

Mohammad Salahinezhad^{a,*}, Hamed Tahami^b, Seyedeh Elham Kamali^c, Ehsan Saedpanah^d

^a Department of Engineering, Faculty of Mechanical Engineering, Islamic Azad University Sari Branch, Iran

^b School of Industrial and Information Engineering, Politecnico di Milano, 20133, Milan, Italy

^c Department of Mechanical and Industrial Engineering, University of Newhaven, Newhaven, United States

^d Faculty of Mechanical Engineering, Tarbiat Modares University, PO Box 14115-143, Tehran, Iran

ARTICLE INFO

Keywords:

Solar collector
Transient simulation
Multi-objective optimization
Life cycle assessment
Hybrid energy systems
Cooling systems

ABSTRACT

This study presents a novel optimization strategy, called the transient simulation-response surface method-life cycle assessment (TRS-RSM-LCA), for the optimal design of a trigeneration or combined cooling, heat, and power (CCHP) system incorporating solar collectors and hydrogen generation subsystems. The TRS-RSM-LCA aims to achieve the optimal energy, economic, and environmental (3E) design of the proposed hybrid energy system throughout its lifespan. The optimization study considers the solar collector area and fuel cell capacity as numeric factors, and the solar collector type and cooling system type as categorical factors. The study investigates the interactions among these design parameters and their impact on the 3E indicators. Results demonstrate that the optimized CCHP system significantly outperforms the base system. The optimized system achieves a 58 % reduction in net electricity consumption (7581.6 kWh) compared to the base system (18,051.6 kWh). Moreover, it exhibits a 38.2 % decrease in net natural gas consumption, a 39.5 % reduction in life cycle cost, and a 46.3 % reduction in 100-year global warming potential. The comparative analysis findings provide valuable insights for decision-makers, enabling them to evaluate the overall system performance and understand the trade-offs involved in optimizing multiple factors.

Nomenclature

A	area (m ²)	t	time (s)
a_0	intercept efficiency	tc_n	coefficients for the overall cooling capacity
a_1	first – order efficiency	th_n	coefficients for the HP heating capacity
a_2	second – order efficiency	T	temperature (°C)
a_r	radiative effectiveness for reference gas	TA	transmittance – absorptance product for PVT
a_x	radiative effectiveness	TH	time horizon (year)
AOC	annual operating cost (US\$)	U	voltage (V)
bf_n	polynomial coefficients for bypass fraction of HP	U_L	overall thermal loss coefficient (kJ.h ⁻¹ .m ⁻² .K ⁻¹)
C	capacitance of the collector (kJ.K ⁻¹)	$U_{L/T}$	quadratic loss coefficient
C_p	specific heat capacity (kJ.kg ⁻¹ .K ⁻¹)	$\dot{V}_{H_2}$	total hydrogen consumption (m ³)

(continued on next page)

* Corresponding author.

E-mail addresses: Mohammad.Salahinezhad70@gmail.com (M. Salahinezhad), Hamed.tahami@mail.polimi.it (H. Tahami), Skama12@unh.newhaven.edu (S.E. Kamali), ehsansaedpanah@gmail.com (E. Saedpanah).

<https://doi.org/10.1016/j.renene.2024.122211>

Received 4 August 2024; Received in revised form 1 November 2024; Accepted 18 December 2024

Available online 23 December 2024

0960-1481/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(continued)

CD	composite desirability	x	axial coordinate (m)
COP	coefficient of performance	$x(t)$	transient reduction in abundance of system's instant release
d	discount rate	y	response
D	electricity demand(kWh)	z	factor
de	desirability	Abbreviations	
e_m	emissivity of the PVT surface	AC	absorption chiller
f	operation status	3E	energy, economic, and environmental
f_{bypass}	bypass fraction	CCD	central composite design
f_{EIR}	rated ratio of energy input in HP	CCHP	combined cooling, heat, and power
f_{sens}	rated sensible capacity	DOE	design of experiments
$f_{tot,c}$	rated overall cooling capacity	ETC	evacuated tube collector
f_{heat}	rated heating capacity	GWP	global warming potential
F'	collector fin efficiency	HP	heat pump
G	electricity generation (kWh)	LCA	life cycle assessment
G_T	total solar radiation incident on PVT	PTC	parabolic trough collector
GWP100	100 – year global warming potential	PVT	photovoltaic/thermal
h	enthalpy (kJ.kg^{-1})	RSM	response surface method
hp_n	coefficients EIR of HP in heating	TRS – RSM – LCA	transient simulation – response surface method – life cycle assessment
i	inflation rate	TESS	thermal energy system specialists
IAM	incidence angle modifier	Scripts	
I_C	initial investment cost (US\$)	a	air
I_T	total radiation incident	b	building
k	thermal conductivity ($\text{W.m}^{-1}.\text{K}^{-1}$)	cm	compressor
L	PVT collector length (m)	cs	cooling system
LCC	life cycle cost (US\$)	cw	cooling water stream
LHV	lower heating value (kJ.m^{-3})	db	dry bulb
$\dot{m}$	mass flow rate (kg.s^{-1})	f	fluid
n_L	lifespan (years)	fa	fan
$n_{c,ser}$	number of cells in series	FC	Fuel cell
$n_{m,ser}$	number of modules in series	H	heat transfer fluid
$\dot{n}_{H2}$	hydrogen production (m^3)	in	inlet
NE	net electricity consumption (kWh.year^{-1})	hw	hot water stream fluid
NN	natural gas consumption ($\text{m}^3.\text{year}^{-1}$)	out	outlet
N_f	number of factors	pu	pump
N_r	number of responses	set	setpoint
N_t	number of time steps	wb	wet bulb
N_{tubes}	number tubes in PVT collector	Greek symbols	
PWF	present worth factor	Δt	time step
Q_{gen}	heat generation(kW)	φ	coefficients obtained from regression
$\dot{q}$	HP cooling rate(kJ.h^{-1})	η	efficiency
$[r(t)]$	transient reduction in abundance of system's instant release for reference gas	η_f	Faraday efficiency
R_I	resale(US\$)	$(\tau\alpha)_n$	product of cover transmittance and absorber absorptance at normal incidence
SC_n	coefficients for sensible cooling capacity	θ	angle of incidence

1. Introduction

Global energy consumption continues to rise significantly, driven by population growth and industrialization, resulting in increased greenhouse gas emissions and intensifying the urgency for sustainable energy solutions [1]. Combined cooling, heat, and power (CCHP) systems, also known as trigeneration systems, are essential for the optimal integration of renewable energy sources in residential apartment communities [2]. These innovative systems combine electricity generation, heating, and cooling into a single process, utilizing a variety of energy sources such as solar power, wind energy, or biomass [3]. The incorporation of renewable energy sources into CCHP systems reduces reliance on fossil fuels [4]. The considerable energy consumption of such areas emphasizes the

need to research and deploy CCHP systems in residential communities. Residential buildings, for example, account for over 40 % of total global energy usage [5]. By optimizing energy production, storage, and distribution, CCHP systems can significantly contribute to reducing energy costs, enhancing energy efficiency, and mitigating environmental impacts for residential apartment communities [6].

Complex hybrid systems require careful analysis and fine-tuning because of their numerous interdependent components and varied energy sources [7]. Researchers can determine the best types of equipment, their capacities, and control schemes through optimization, which maximizes energy use and minimizes energy waste [8]. For instance, Ren et al. [9] proposes hybrid CCHP system, which is equipped with a gas turbine, water storage tank, absorption unit, ground source heat pump, photovoltaic panels (PV), solar thermal collectors, batteries, and photovoltaic/thermal collectors (PVT). The evaluation is done on two

distinct solar energy frameworks: framework A employs solar energy conversion into thermal and power through PV and solar thermal collector, accordingly, while framework B employs solar energy conversion into thermal and power through PVT. They employed technique for order performance by similarity to ideal solution (TOPSIS) approach as a reference for decision-making when choosing the best setup. According to their findings, system A that operates using the following electric load (FEL) method performs better. Ghorbani et al. [10] investigated a geothermal energy heat recovery-based power generation unit. They used the Kalina cycle, ejector refrigeration cycle, and organic Rankine cycle to form the three modules of the power generation unit. Additionally, the goal criteria including net generated power, overall investment cost, and energy efficiency are selected using the TOPSIS multi-objective optimization approach. According to their findings, 287.4 K is the ideal level according to the evaporator temperature at the geothermal thermal source temperature of 480 K. The objective of Wang et al. [11] was to analyze the thermodynamic performance and optimize the configurations of a hybrid CCHP system that combines solar energy and natural gas. The analysis revealed that integrating solar PV into the CCHP system improves exergy efficiency, while the integration of a solar heat collector enhances the energy efficiency. Using the primary energy saving ratio (PESR) and yearly cost saving ratio (CSR) as the goal of optimization study, Song et al. [12] performs an optimization study of a CCHP system integrated with solar energy. Their optimization outcomes indicate a contradiction among fuel economy and cost reduction. The average PESR rises to around 37.9 %, and the average CSR rises to 25 %, approximately, when the constrained optimum point is chosen.

A statistical method for optimizing hybrid energy systems is called response surface method (RSM). To predict the relationship between input variables and intended responses, RSM builds a mathematical model [13]. In order to find the best solutions for hybrid energy systems, RSM makes it possible to explore the design space systematically [14]. Saedpanah et al. [15] developed an optimization method to achieve the best possible layout of a hybrid air conditioning system, which consists of a thermal energy storage, a desiccant wheel, PVT panels, ground source heat exchanger, and absorption chiller. Their findings demonstrated that the system's yearly natural gas and electrical power usage, applying the optimal design discovered by RSM, decreased by 51.8 % and 14.6 %, respectively. Mao et al. [16] used RSM to accelerate the computer modeling process to estimate the interior temperature environment. They compared the findings from the numerical technique (RSM method) alongside the computational fluid dynamics (CFD) approach in order to examine the energy efficiency and thermal comfort conditions of a bedroom tac unit. Tooryan et al. [17] investigated the optimal design and energy management of a hybrid microgrid system with distributed energy resources. The study proposes an approach to minimize operational costs by incorporating photovoltaic arrays, battery energy storage, fuel cells, thermal energy storage, and boilers. They explored the utilization of residential and municipal waste for electricity generation, along with load growth considerations. They employed Particle Swarm Optimization Algorithm to determine the best energy management strategy and resource capacity. Their results showed reductions in natural gas consumption by fuel cells (6.2 % with residential waste and 26.7 % with residential and municipal waste) and a significant decrease in CO₂ emissions (46.8 %) through the utilization of fuel cell heat. Nedaei et al. [18] studied the design and optimization of a hybrid power system. The system's objective is to produce electricity from wind power. If there is excess electricity, it may be utilized to produce hydrogen on a small scale or transferred to the grid for medium- or large-scale electrification. Their results showed that the estimated daily wind energy production transmitted to the grid is around 2000 kWh. By adjusting the hybrid energy system and considering various design coefficients such as electricity price, hydrogen production, hydrogen price, and electricity transmitted to the grid, the study finds that the net monthly revenues can reach a maximum level of €1974.87.

The need for sustainable and efficient energy systems has intensified

due to rising environmental concerns, the increasing demand for energy, and the need for economically viable solutions in urban areas. Combined cooling, heating, and power (CCHP) systems, which can integrate renewable energy sources, present a promising avenue for meeting these demands, especially in dense residential communities. However, optimizing these systems remains a complex challenge that requires balancing multiple objectives—energy efficiency, economic feasibility, and environmental sustainability—each influenced by various system components and configurations.

Current literature reveals significant gaps in understanding the optimization of hybrid CCHP systems for maximal performance across these three pillars. Specifically,

- Research Gap #1: Existing studies primarily focus on multi-objective optimization for CCHP systems, but often neglect to explore different weighting schemes or decision-making techniques that could prioritize energy, economic, or environmental objectives more effectively. Analyzing these trade-offs with alternative approaches could yield valuable insights into the performance boundaries and potential compromises in CCHP system design.
- Research Gap #2: Previous studies have not yet examined the simultaneous optimization of solar collector type and cooling system type within hybrid CCHP systems. While some research addresses these elements in isolation, a holistic approach that considers their combined impact is essential to advancing the energy efficiency and sustainability of hybrid CCHP systems.
- Research Gap #3: The life cycle assessment (LCA) of hybrid CCHP systems—accounting for environmental impacts such as embodied energy, emissions, and waste generation throughout the system's lifecycle—remains underexplored. A comprehensive LCA could provide critical insights into the long-term sustainability of these systems and inform environmentally responsible design choices.

To address these gaps, this study introduces a novel transient simulation-response surface method-life cycle assessment (TRS-RSM-LCA) framework for the hybrid CCHP system, designed to optimize component configuration while balancing energy, economic, and environmental (3E) objectives. This approach not only simulates time-dependent heat transfer and thermodynamics but also provides a nuanced understanding of the system's environmental impacts across its lifecycle. Using TRNSYS for transient simulations, RSM for design of experiments, and LCA for evaluating lifecycle impacts, this strategy aims to identify an optimal system configuration for a residential apartment community. A subsequent trade-off analysis further contextualizes these findings, offering practical insights for stakeholders on how best to balance the 3E objectives in real-world applications.

2. Materials and methods

This section presents the description of the system, modeling, and optimization methodology used in this study.

2.1. Description of system

The hybrid system described consists of five major subsystems: the hydrogen generation subsystem, hot water storage subsystem, solar subsystem, cooling subsystem, and controller subsystem. Each subsystem plays a crucial role in the overall operation and efficiency of the system.

The hydrogen generation subsystem comprises an electrolyzer, a compressor, a hydrogen storage tank, and a fuel cell system. This subsystem is responsible for generating electricity through the process of electrolysis, where water is split into hydrogen and oxygen. The generated hydrogen is stored in a tank, and a fuel cell converts it back into electricity when needed. Additionally, the fuel cell can utilize the heat produced during its operation to heat a stream of water, which is

then returned to the hot water storage tank.

The hot water storage subsystem includes a hot water storage tank and a backup boiler. The storage tank has multiple inlets and outlets, with each serving a specific purpose. The first inlet and outlet are connected to the fuel cell subsystem, allowing the transfer of heated water between the fuel cell and the storage tank. The second inlet and outlet are linked to the solar subsystem, enabling the transfer of solar-heated water to the storage tank. The third inlet and outlet are associated with the cooling subsystem (specifically, if level 2 of the cooling subsystem is chosen), allowing the transfer of hot water from the storage tank to fulfill the cooling demand. Lastly, the fourth inlet and outlet are connected to the backup boiler, which is responsible for increasing the temperature of the tank water to meet the hot water consumption needs of the apartment complex.

The solar subsystem offers three levels in the decision-making process: level 1 utilizes an evacuated tube collector (ETC), level 2 employs a parabolic trough collector (PTC), and level 3 incorporates a photovoltaic/thermal collector (PVT). The solar subsystem's primary function is to raise the water temperature in the hot water storage tank using the available solar radiation. The selection of the appropriate solar system type is determined through an optimization methodology, which is discussed in detail in the paper.

The cooling subsystem is designed to meet the cooling demand of the apartment complex. It offers two levels: level 1 utilizes an air-source heat pump (HP), and level 2 incorporates an absorption chiller (AC). The level 1 HP system operates by extracting heat from the ambient outdoor air and transferring it to the indoor space, effectively providing cooling. The HP utilizes a refrigeration cycle, driven by electricity, to facilitate this heat transfer process. By leveraging the temperature difference between the indoor and outdoor air, the HP can provide efficient cooling without the need for a direct thermal energy input. If the level 2 AC system is chosen during the decision-making process, the required thermal energy for the AC generator is provided by extracting a hot water stream from the hot water storage tank. This integration enables the utilization of waste heat from the system to fulfill the cooling requirements, further enhancing the overall energy efficiency. The key difference between the level 1 HP system and the level 2 AC system is the source of thermal energy used to drive the cooling process. While the level 2 AC system relies on the waste heat extracted from the hot water storage tank, the level 1 HP system uses electricity to power the compressor and facilitate the heat transfer. The selection between the level 1 HP and level 2 AC systems is determined through the optimization methodology described in the paper. This decision-making process takes into account factors such as the available solar thermal energy, the cooling demand, the system efficiency, and the overall energy consumption and cost implications. The optimization aims to select the most appropriate cooling subsystem configuration to maximize the energy efficiency, cost-effectiveness, and environmental sustainability of the overall system.

The controller subsystem plays a crucial role in managing the electricity consumption and generation of the entire hybrid system. It is responsible for monitoring and controlling the electricity flow based on the electricity generated by the fuel cell subsystem and level 3 of the solar subsystem, as well as the electricity demand of the apartment complex. The controller subsystem makes decisions regarding the amount of electricity available for sale to the grid or the amount of electricity that needs to be purchased from the grid.

2.2. System modeling

To simulate an advanced hybrid CCHP system that integrates solar energy and hydrogen generation technologies, Transient System Simulation Tool (TRNSYS) and the Fortran programming language are used. The TRNSYS and Thermal Energy System Specialists (TESS) libraries offer a wide range of Fortran-coded components (i.e., Types) that are effectively integrated with TRNSYS to model the proposed system.

TRNSYS had been chosen as the main modeling program due to its established credibility in the research community [19]. Moreover, certain experimental studies have verified the dependability of the Fortran codes used in TRNSYS Types [20–26].

The performance characteristics of the evacuated tube solar collector (ETC) are determined by utilizing Eq. (1) to calculate the thermal efficiency [27]. This equation is implemented within Type 71 of the TRNSYS software, where this component effectively models the behavior of ETC by employing a standard quadratic efficiency equation. Type 1245 of the TESS library is employed to determine the behavior of the parabolic trough solar collector (PTC). This particular component effectively models a medium-temperature non-tracking concentrating parabolic solar collector, taking into account the influence of collector mass on performance. PTC collectors consist of an evacuated tube containing an absorber and heat transfer fluid, with an external reflective concentrator located at the back. The model calculates the collector fin efficiency (F) to determine the collector loss coefficient and the transmittance-absorptance product at normal incidence, as depicted in Eq. (2) [28]. Type 560 of the TRNSYS software, utilized in various studies such as [15,29,30], is specifically designed to model PVT panels. This model enables the calculation of both the fluid temperature at the collector outlet and the generated power by utilizing Eqs. (3) and (4) [31]. The Type 560 model employs linear variables to establish the correlation between photovoltaic effectiveness, cell temperature, and incident solar radiation. However, it does not incorporate degradation factors into its calculations.

Type 966 from the TESS library is employed to analyze the characteristics of a heat pump (HP). This particular model utilizes a set of empirical coefficients to accurately represent the behavior of an air-to-air heat pump [32]. By considering the entering evaporator air wet bulb temperature and the condenser air dry bulb temperature, Type 966 calculates various parameters such as total and sensible capacity, energy efficiency ratio (EIR), and bypass fraction at the current inlet conditions. Eqs. (5)(8) [28] are incorporated within Type 966 to facilitate the modeling of the heat pump (HP). To model the behavior of the absorption chiller (AC), the study utilizes Type 107 of the TRNSYS software. Within this component, Eqs. (9) and (10) [32] are used to determine the outlet temperature of the hot water stream from the generator and the cooling water stream, respectively.

In order to simulate fuel cells, Type 173 is utilized, incorporating Eq. (11)(17) from Refs. [27,33]. For the accurate simulation of the electrolyzer and its hydrogen production, Type 160 of the TRNSYS software is employed, with Eq. (18) defining the ultimate rate of hydrogen production [27]. It is important to acknowledge that the density of connected cells directly impacts the hydrogen production rate of the electrolyzer.

The governing equations for other components such as backup boiler, pumps, fans, compressor, and storage tanks are not presented here due to the extensive number of equations involved. However, these equations are fully considered to ensure accurate simulations, and they can be found in Ref. [27].

The following main assumptions are incorporated into the applied mathematical model, based on the referenced resources.

- The radial conductive thermal resistance of the pipes is considered imperceptible in the model [19].
- Radial heat conduction is not taken into account in the model [34].
- In the electrolyzer subsystem Hydrogen and air are assumed to be ideal gases [29].
- The water used as heat transfer fluid is considered to be an incompressible fluid [27,35].
- The model for the evacuated tube solar collector assumes that the relationship between efficiency and the ratio of the temperature difference between the fluid and ambient temperature to the radiation curve can be represented by a quadratic equation [28].

- The model for a parabolic trough collector assumes that the heat loss to the ambient environment is proportional to the square of the temperature difference between the fluid and ambient air, with a quadratic loss coefficient [28].

$$\eta_{thermal} = a_0 - a_1 \frac{(\Delta T)}{I_T} - a_2 \frac{(\Delta T)^2}{I_T} \quad (1)$$

$$C \frac{dT}{dt} = F((\tau\alpha)_n IAM I_T - AU_L(T - T_a) - AU_{L/T}(T - T_a)|(T - T_a)|) - \dot{m}C_p(T - T_{in}) \quad (2)$$

$$T_{H,out} = \left(T_{H,in} + \frac{e_m}{\kappa}\right) \exp\left(\frac{N_{tubes}}{\dot{m}_H C_H} \frac{\kappa}{\theta} L\right) - \frac{e_m}{\kappa} \quad (3)$$

$$P_{PVT} = (TA)_n \cdot IAM \cdot G_T \cdot A \cdot \eta_{PV/T} \quad (4)$$

$$\begin{cases} f_{totCool} = tc_1 + tc_2 T_{wb,in} + tc_3 T_{wb,in}^2 + tc_4 T_{db,amb} + tc_5 T_{db,amb}^2 + tc_6 T_{wb,in} T_{db,amb} \\ f_{sensCool} = sc_1 + sc_2 T_{wb,in} + sc_3 T_{wb,in}^2 + sc_4 T_{db,amb} + sc_5 T_{db,amb}^2 + sc_6 T_{wb,in} T_{db,amb} \\ f_{EIR,c} = cp_1 + cp_2 T_{wb,in} + cp_3 T_{wb,in}^2 + cp_4 T_{db,amb} + cp_5 T_{db,amb}^2 + cp_6 T_{wb,in} T_{db,amb} \\ f_{Bypass} = bf_1 + bf_2 T_{wb,in} + bf_3 T_{wb,in}^2 + bf_4 T_{db,amb} + bf_5 T_{db,amb}^2 + bf_6 T_{wb,in} T_{db,amb} \end{cases} \quad (5)$$

$$\begin{cases} f_{heat} = th_1 + th_2 T_{wb,in} + th_3 T_{wb,in}^2 + th_4 T_{db,amb} + th_5 T_{db,amb}^2 + th_6 T_{wb,in} T_{db,amb} \\ f_{EIR,h} = hp_1 + hp_2 T_{wb,in} + hp_3 T_{wb,in}^2 + hp_4 T_{db,amb} + hp_5 T_{db,amb}^2 + hp_6 T_{wb,in} T_{db,amb} \end{cases} \quad (6)$$

$$\begin{cases} h_{coilOut} = h_{airIn} - \frac{\dot{q}_{tot}}{\dot{m}_{air}(1 - f_{bypass})} \\ T_{coilOut} = T_{dbIn} - \frac{\dot{q}_{sens}}{\dot{m}_{air} C_{air}(1 - f_{bypass})} \end{cases} \quad (7)$$

$$\begin{cases} h_{airOut} = f_{bypass} h_{airIn} + (1 - f_{bypass}) h_{coilOut} \\ \omega_{airOut} = f_{bypass} \omega_{airIn} + (1 - f_{bypass}) \omega_{coilOut} \end{cases} \quad (8)$$

$$T_{hw,out} = T_{hw,out} - \frac{\dot{Q}_{hw}}{\dot{m}_{hw} c_{hw}} \quad (9)$$

$$T_{cw,out} = T_{cw,out} - \frac{\dot{Q}_{cw}}{\dot{m}_{cw} c_{cw}} \quad (10)$$

$$U_{cell} = \frac{U_{mod}}{n_{c,ser}} \quad (11)$$

$$U_{mod} = U_0 - b \log(I_{stack}) - R_{ohm} I_{stack} \quad (12)$$

$$U_{stack} = n_{m,ser} U_{mod} \quad (13)$$

$$E_{stack} = U_{stack} I_{stack} \quad (14)$$

$$\eta_E = \frac{U_{cell} \cdot n_e \cdot F}{\Delta H_2} \quad (15)$$

$$\dot{V}_{H_2} = \frac{n_{c,ser} n_{m,ser} I_{FC}}{z F \rho_{gas}} S_{H_2} \eta_F \quad (16)$$

$$Q_{gen} = E_{stack} \left(\frac{1 - \eta_E}{\eta_E} \right) \quad (17)$$

$$\dot{n}_{H_2} = \eta_f N_{cells} \frac{I_{ely}}{n_F} \quad (18)$$

2.3. Optimization methodology

The primary objective of this study is to develop an optimization strategy for a hybrid CCHP system that integrates solar energy and hydrogen generation (see Fig. 1). The aim is to achieve optimal energy, economic, and environmental (3E) performance throughout the system's lifespan. Nevertheless, traditional optimization methods are computationally expensive for long simulation periods with time steps

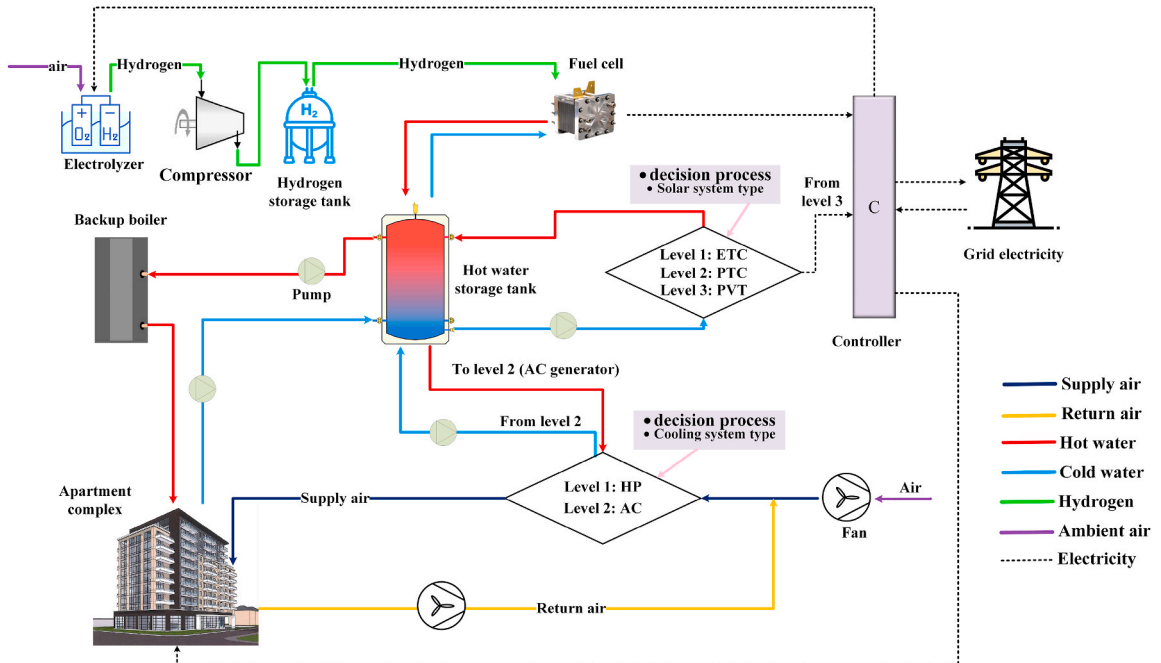


Fig. 1. The schematic of the studied hybrid system.

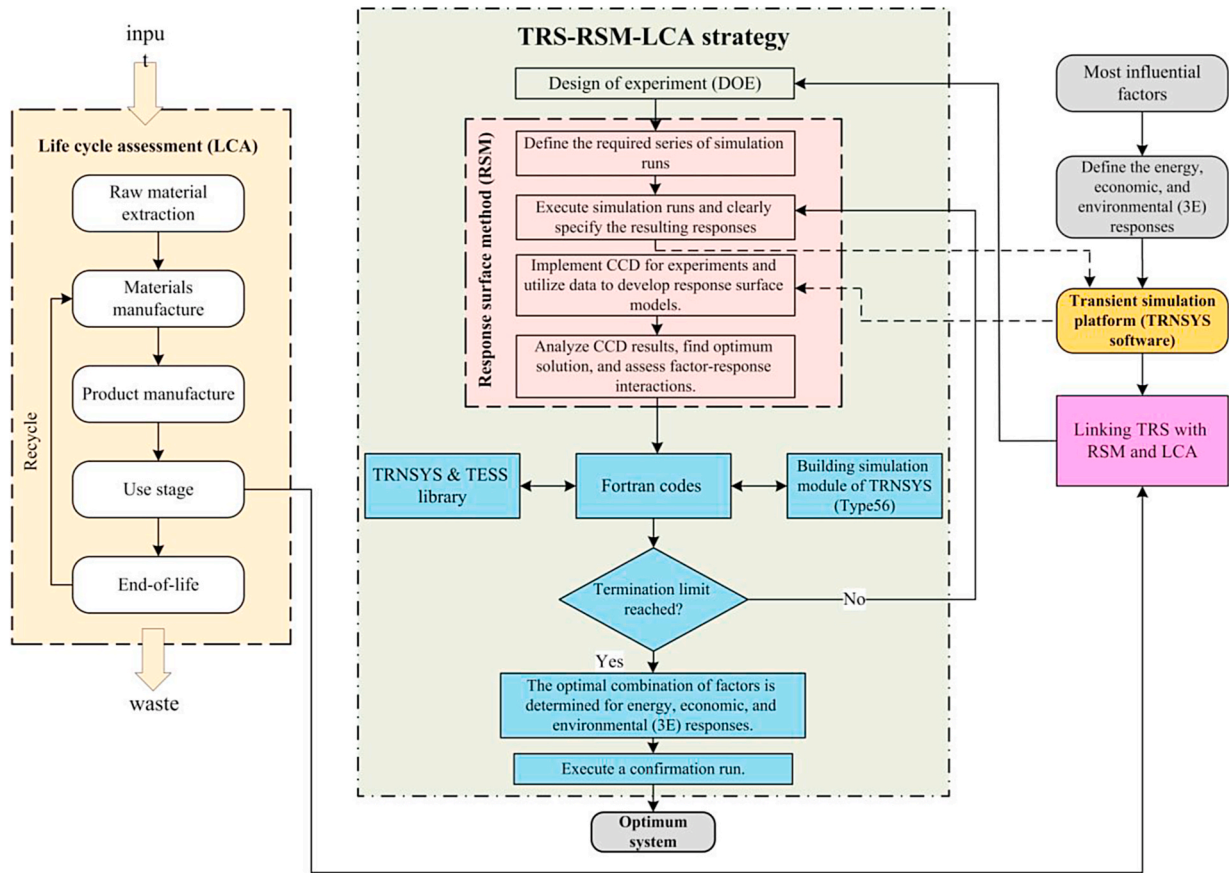


Fig. 2. In-depth description of the optimization strategy.

Table 1

The selected factors for the optimization procedure.

Factor	A	B	C	D
Name	Solar collector area	Fuel cell capacity	Solar collector type	Cooling system type
Factor type	Numeric	Numeric	Categoric	Categoric
Unit	m ²	kW	–	–
Minimum	0	0	–	–
Maximum	250	100	–	–
Levels	–	–	Level 1 : Evacuated tube collector (ETC) Level 2 : Parabolic trough collector (PTC) Level 3 : Photovoltaic/thermal collector (PVT)	Level 1 : Heat pump (HP) Level 2 : Absorption chiller (AC)

ranging from a few seconds to a few hours. To address this, a novel optimization strategy, referred to as TRS-RSM-LCA, is proposed and depicted in Fig. 2. It combines transient simulation (TRS), response surface method (RSM), and life cycle assessment (LCA). The strategy involves a DOE to classify simulation cases based on selected factors and constraints. TRS and LCA results are used to create a response surface model for predicting 3E outcomes over the system's lifespan. The TRS-RSM-LCA approach significantly reduces computational costs compared to other methods by minimizing the number of simulation cases required.

2.4. RSM detail

RSM is a statistical and mathematical technique for optimizing and evaluating complex systems. RSM models and explores the response

surface using a combination of numerical simulation, regression analysis, and graphical representation [36]. RSM entails creating a set of simulation cases to collect data at various levels of the input variables [37]. The response surface, which depicts the relationship between the input variables and the output response, is represented by this model [38].

The response surface model is developed using central composite design (CCD) and a second-order equation (Eq. (19)) [39]. The objective of this study is to optimize the 3E indicators of the proposed system collectively, employing the desirability function technique. The desirability functions for each response are combined to calculate the composite desirability (CD) (Eq. (20)) [40]. A CD value close to 1 signifies a desirable outcome in the multi-response optimization process.

Table 2

The details of selected residential apartments [15,46–48].

Parameter (unit)	Value
Unit area	97.1 (m ²)
Total conditioned area (all units)	4660.8 (m ²)
Number of units	48
Units per flat	4
Occupants of each unit	5 (people)
External U value	1.3 (W.m ⁻² .K ⁻¹)
Internal U value	1.2 (W.m ⁻² .K ⁻¹)
Overall U value of floor	0.318 (W.m ⁻² .K ⁻¹)
Overall U value of windows	3.1 (W.m ⁻² .K ⁻¹)
Ceiling height	2.8 (m)
Area of each window	1.7 (m ²)
Wall to window ratio	0.5
Internal shading factors	0.8
External shading factors	0.2

$$y = \varphi_0 + \sum_{i=1}^{N_f} \varphi_i z_i + \sum_{i=1}^{N_f} \varphi_{ii} z_i^2 + \sum_{i < j=2}^{N_f} \sum_{j=2}^{N_f} \varphi_{ij} z_i z_j \quad (19)$$

$$CD = (de_1 \times de_2 \times \dots \times de_{N_r})^{\frac{1}{N_r}} = \left(\prod_{i=1}^{N_r} de_i \right)^{\frac{1}{N_r}} \quad (20)$$

2.5. Design factors of the optimization strategy

Four independent factors are considered as key design variables in

this study: the solar collector area, fuel cell capacity, solar collector type, and cooling system type. The solar collector area, and fuel cell capacity are the numeric factors, while solar collector type, and cooling system type are the categoric factors. Table 1 illustrates the variations of these factors and the associated constraints. The upper limits are determined based on installation space constraints. Factor C, the solar collector type, consists of three levels: level 1 represents evacuated tube collector (ETC), level 2 represents parabolic trough collector (PTC), and level 3 represents photovoltaic/thermal collector (PVT). Furthermore, factor D, the cooling system type, has two levels: level 1 corresponds to heat pump (HP) and level 2 corresponds to absorption chiller (AC). To optimize the process, a total of 78 simulation cases are recommended by CCD for the selected factors. The Design-Expert platform [41] is utilized for conducting the necessary calculations using the response surface method and DOE (see Table 2).

The following assumptions have been considered in the optimization process.

1. **Linearity and Curvature of Relationships:** It is assumed that the relationships between input variables (design factors) and the response (3E indicators) can be represented adequately by a second-order polynomial model. This assumption allows the use of a quadratic model for accurate response prediction and optimization.
2. **Fixed Bounds of Factors:** The maximum and minimum values for the numeric factors (solar collector area and fuel cell capacity) are based on practical constraints, such as available installation space and component capabilities. These boundaries are assumed to represent realistic limits within which the system operates effectively.

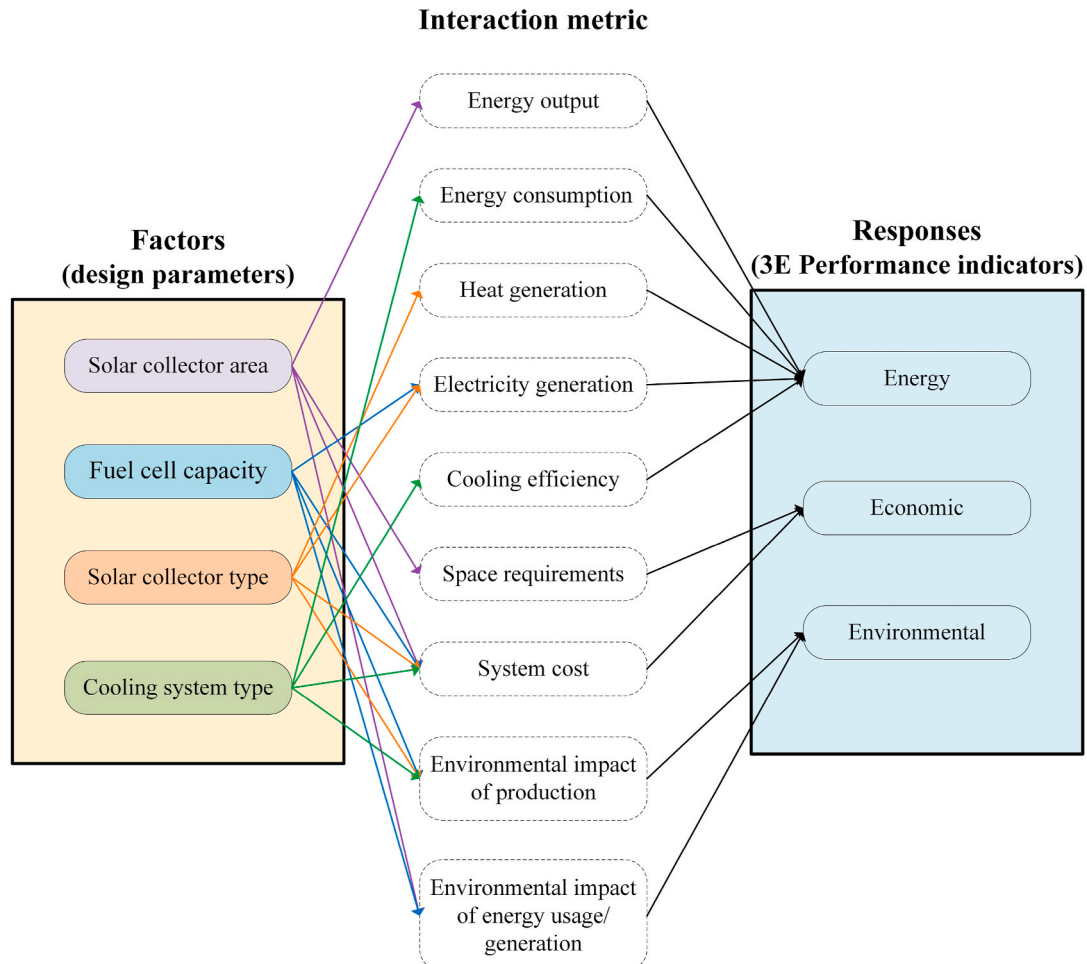


Fig. 3. The flowchart of interaction between design factors and 3E performance indicators in the optimization strategy.

3. Independence of Factors: Each of the design factors (solar collector area, fuel cell capacity, solar collector type, and cooling system type) is considered to be independent of the others. This means that interactions among factors are minimal or manageable within the design space, allowing the response surface model to focus on main effects and selected interactions.
4. Desirability Function Validity: It is assumed that the desirability function adequately represents the collective optimization of the 3E indicators. The composite desirability (CD) value close to 1 indicates optimal conditions across multiple responses, thus assuming that all objectives can be meaningfully balanced.
5. Categorical Factor Levels Represent Feasible Technologies: The categorical levels for solar collector and cooling system types are chosen based on technologies that are readily available and feasible for integration into CCHP systems. It is assumed that each level provides distinctive performance characteristics relevant to the optimization objectives.

2.6. 3E responses of the optimization strategy

In this research, a comprehensive analysis is conducted, considering four distinct parameters: two energy indicators, an economic indicator, and an environmental indicator. The net electricity consumption (NE) and net natural gas consumption (NN) are two of the energy parameters under examination. These values are calculated using Eqs. (21) and (22), respectively [15,34].

$$NE = \frac{1}{3600} \sum_{i=1}^{N_t} (D_b \cdot f_b + D_{cs} \cdot f_{cs} + D_p \cdot f_p + D_f \cdot f_f + D_{cm} \cdot f_{cm} - G_{PVT} \cdot S_{PVT} - G_{FC} \cdot f_{FC}) \Delta t \quad (21)$$

$$NN = \frac{\sum_{i=1}^{N_t} (\dot{m}_f c_f (T_{set} - T_{f,in}) f_{boiler})}{LHV \eta_{boiler}} \quad (22)$$

The life cycle cost (LCC) indicator is utilized to evaluate the system's economic response. The total cost of a system is referred to as its LCC analysis, which involves evaluating and aggregating various cost components, such as initial investment, operating costs, maintenance and repair costs, energy consumption, replacement or upgrade costs, and any potential salvage value or residual costs [42]:

$$LCC = I_c + PWF \cdot AOC - R_t; PWF = \begin{cases} \frac{1}{d-i} \left[1 - \left(\frac{i+1}{d+1} \right)^{n_L} \right] & \text{if } i \neq d \\ \frac{n_L}{i+1} & \text{if } i = d \end{cases} \quad (23)$$

As stated before, the methodology known as LCA, is used to assess how the proposed hybrid system will affect the environment over the course of its lifespan. From the extraction of raw materials to manufacture, use, and disposal, it takes into account every step. LCA aids in the identification and measurement of environmental costs such as waste production, emissions, energy use, and resource depletion. In LCA, the 100-year global warming potential (GWP100) is a metric used to measure the possible effect of greenhouse gas emissions on global warming over a 100-year period [43]. It is a measurement of the relative contribution of a specific greenhouse gas to global warming during a given time period when compared to carbon dioxide (CO₂) [44]. Based on its warming potential in relation to CO₂, each greenhouse gas is given a numerical value by GWP100. With a global warming potential (GWP) of 1, carbon dioxide serves as the reference gas. Other greenhouse gases with considerably higher warming potentials and GWP values include nitrous oxide (N₂O) and methane (CH₄). The impact of these emissions on global warming over a 100-year period is then determined by multiplying them by their corresponding GWP100 values [45]. The

CO₂-equivalents used to define the total GWP100 value enable a standard comparison of the overall climate change impact of various items or activities [15]:

$$GWP(x) = \frac{\int_0^{TH} a_x \cdot [x(t)] dt}{\int_0^{TH} a_r \cdot [r(t)] dt} \quad (24)$$

Subsequently, the OpenLCA software is employed to perform computations utilizing LCA methodology.

Fig. 3 illustrates the interactions between the design parameters—solar collector area, fuel cell capacity, solar collector type, and cooling system type—and the 3E performance indicators: energy, economic, and environmental outcomes. The flowchart identifies how each design factor influences key interaction metrics, such as energy output, cooling efficiency, and system cost. These metrics ultimately affect the overall 3E performance, guiding the optimization process by clarifying the impact of individual design choices on the system's holistic performance. This visual representation enhances the understanding of complex parameter interdependencies within the proposed optimization framework.

2.7. Overview of the TRS-RSM-LCA methodology

The primary objective of this study is to develop an optimization strategy for a hybrid CCHP system, integrating solar energy and hydrogen generation to maximize energy, economic, and environmental (3E) performance over the system's lifecycle. The TRS-RSM-LCA approach combines three complementary methods.

1. Transient Simulation (TRS): TRS is applied to capture the system's dynamic behavior over time, which is essential for accurately modeling and evaluating performance under real-world operating conditions. The simulations are conducted over time intervals ranging from seconds to hours to account for varying load demands and energy fluctuations, enabling detailed 3E assessment.
2. Response Surface Method (RSM): RSM reduces the computational demands of extensive simulations by creating a response surface model that predicts system behavior. Using central composite design (CCD), RSM facilitates the exploration of multiple scenarios with minimal simulation runs, effectively balancing accuracy with computational efficiency.
3. Life Cycle Assessment (LCA): LCA evaluates the environmental impact of the system throughout its lifespan—from raw material extraction to disposal. This holistic approach provides insight into the environmental costs, such as emissions and resource depletion, of different system configurations and operational strategies.

The integration of these three methods provides a streamlined optimization framework that is both time-efficient and comprehensive, significantly reducing computational demands compared to traditional methods.

2.7.1. Rationale for parameter selection

To achieve optimal 3E performance, four key design factors have been selected: solar collector area, fuel cell capacity, solar collector type, and cooling system type. These factors were chosen based on their direct influence on system performance, including energy efficiency, economic feasibility, and environmental sustainability.

1. Solar Collector Area and Fuel Cell Capacity: These are numeric parameters that determine the system's renewable energy capture and storage potential. A larger solar collector area increases solar energy harnessing, potentially reducing reliance on natural gas (NN), while fuel cell capacity enhances energy storage and the system's ability to meet demand fluctuations.

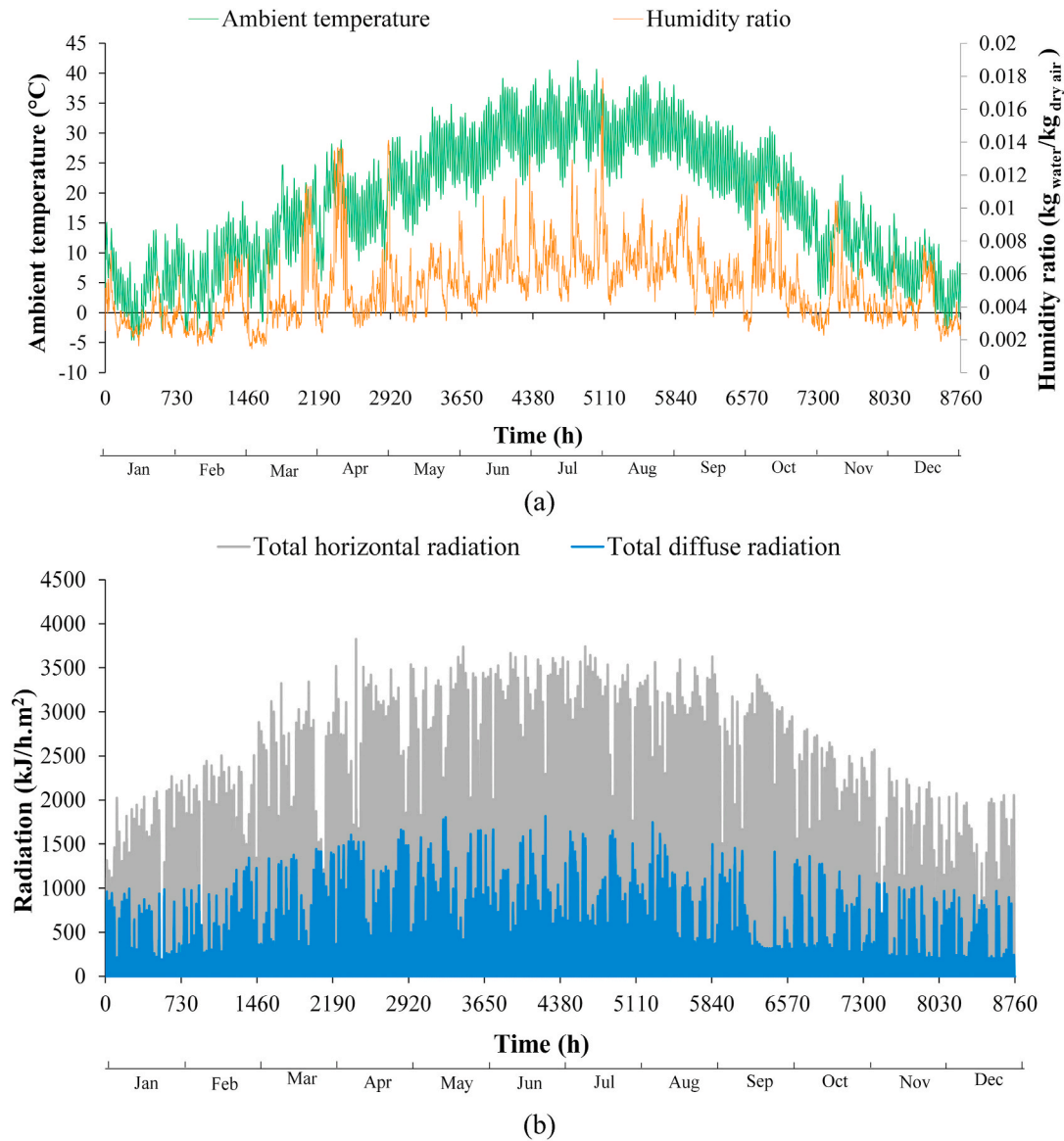


Fig. 4. The climatic factors of the selected case study that impact the building's energy demands (a) ambient temperature and humidity ratio (b) radiation.

Table 3

The employed data used for TRNSYS simulation.

Parameter	Value	Cost	Value
Backup boiler set point temperature	65°C	ETC collector (US\$ per number of ETC collector)	178
Backup boiler fuel	Natural gas	PTC collector (US\$ per number of PTC collector)	98
Solar collector type	ETC/PTC/PVT	PVT collector (US\$ per number of PVT collector)	181
collector bond thermal conductivity (W.m ⁻¹ .K ⁻¹)	45	AC (US\$ /kW)	135.4
collector slope (°)	36	Cooling tower (US\$ /kW of chiller)	12.1
collector azimuth angle (°)	0	HP (US\$ /kW)	305.8
Hot water storage tank volume (L)	90.5 × (number of solar collectors)	ON/OFF differential controller (US\$ per item)	14.2
Solar loop pump flow rate (kg.h ⁻¹)	40 × (area of solar collectors)	Backup boiler (US\$)	1629
Solar collector width (m)	0.992 × (number of solar collectors)	Hot water storage tank (US\$ /L)	0.23
Solar collector length (m)	1.658	Pump (US\$ /kW)	210
Cooling system type	HP/AC	Air fan (US\$ /kW)	230
HP heating energy input ratio	0.332	Cooling/heating coil (US\$ per item)	125
HP cooling energy input ratio	0.256	Fuel cell subsystems (US\$ /kW)	1215
Fuel cell electrode area (cm ²)	100	Electricity purchase price (\$ /kWh)	0.022
Electrolyzer electrode area (cm ²)	0.25	Electricity sale price (\$ /kWh)	0.047
Electrolyzer thermal resistance (K.W ⁻¹)	0.0563	Natural gas purchase price (\$ /m ³)	0.035
system lifespan	30 years	Discount rate (%)	18

2. Solar Collector Type and Cooling System Type: These are categorical parameters that introduce flexibility in design choices, each offering distinct efficiencies, costs, and environmental footprints. Three types of solar collectors—evacuated tube collectors (ETC), parabolic trough collectors (PTC), and photovoltaic/thermal collectors (PVT)—and two types of cooling systems—heat pump (HP) and absorption chiller (AC) have been considered. These options allow us to assess the impact of various technologies on overall system performance.

2.7.2. Implications of parameter choices

Each selected parameter directly affects the 3E outcomes.

1. Solar Collector Area: An increase in collector area enhances renewable energy capture, leading to reduced natural gas consumption but higher installation costs, thus impacting the economic indicator (LCC).
2. Fuel Cell Capacity: Higher capacity improves the system's resilience to peak demand and can lead to greater energy independence, positively affecting both economic and environmental indicators.
3. Solar Collector and Cooling System Types: Different collector and cooling system types have unique efficiencies and environmental profiles. For example, evacuated tube collectors offer high efficiency in low sunlight conditions, while photovoltaic/thermal collectors contribute to both electrical and thermal energy generation, each affecting 3E performance differently.

2.7.3. Advantages of TRS-RSM-LCA

The TRS-RSM-LCA framework allows us to achieve a highly optimized design with fewer simulation runs by leveraging RSM to create predictive models. This strategy also integrates transient simulations with lifecycle assessment, enabling a detailed evaluation of the system's performance over time while minimizing computational resources. The result is a robust, time-efficient approach to optimizing complex systems.

3. Results and discussion

The setup of the simulation, optimization results, and obtained results of the study have been presented in this section.

3.1. Setup of the simulation

This study evaluates the performance of a hybrid CCHP system in a

Table 4

Comparison of the numerical model of this study with experimental data presented by He et al. [52] for system COP of AC and average room temperature provided by AC.

Parameters of validation	Time			
	28-Jul	2-Aug	15-Aug	20-Aug
System COP (Numerical model of this study)	0.22	0.26	0.28	0.39
System COP (Experimental data presented by He et al.)	0.21	0.25	0.30	0.41
Average room temperature (Numerical model of this study)	25.6	26.1	25.6	26.5
Average room temperature (Experimental data presented by He et al.)	24.1	26.8	26.5	25.3
The error of the system COP	4.8 %	4.0 %	6.7 %	4.9 %
The error of the average room temperature	5.9 %	2.5 %	3.6 %	4.7 %

12-story residential building in Tehran, Iran. The building, with 48 units, serves as a representative case with prior references in papers [15, 46–48]. The study analyzes the system's performance over a year, considering Tehran's weather conditions characterized by significant temperature variations. During the cooling and heating seasons, the zones thermostats is set at 26 °C and 21 °C, respectively. The assessment aims to determine the hybrid CCHP system's effectiveness in meeting the building's cooling, heating, and power needs throughout the year.

Fig. 4 illustrates the transient ambient conditions observed throughout the course of a year for the aforementioned case study. The graph depicts the variations in ambient temperature, humidity ratio, total horizontal radiation, and total diffuse radiation, which are climatic factors that impact the building's energy demands.

Table 3 lists the primary TRNSYS simulation parameters as well as the component expenses for the LCC calculations. The manufacturer's websites and regional distributors are used as the primary sources of information about the item's costs. Besides, the Iranian government's information, which has been utilized in multiple studies [49,50], is the basis for the natural gas, electricity purchase, and sale tariff rates.

This study decides not to revalidate TRNSYS components, particularly Types, because the reliability of TRNSYS coded components has been thoroughly validated in prior research through extensive experimental evaluations [20,22–24]. This decision was made strategically to avoid unnecessary repetition, leveraging the well-documented validation from previous studies to instill confidence in the accuracy and dependability of the TRNSYS modules. The study efficiently focuses on

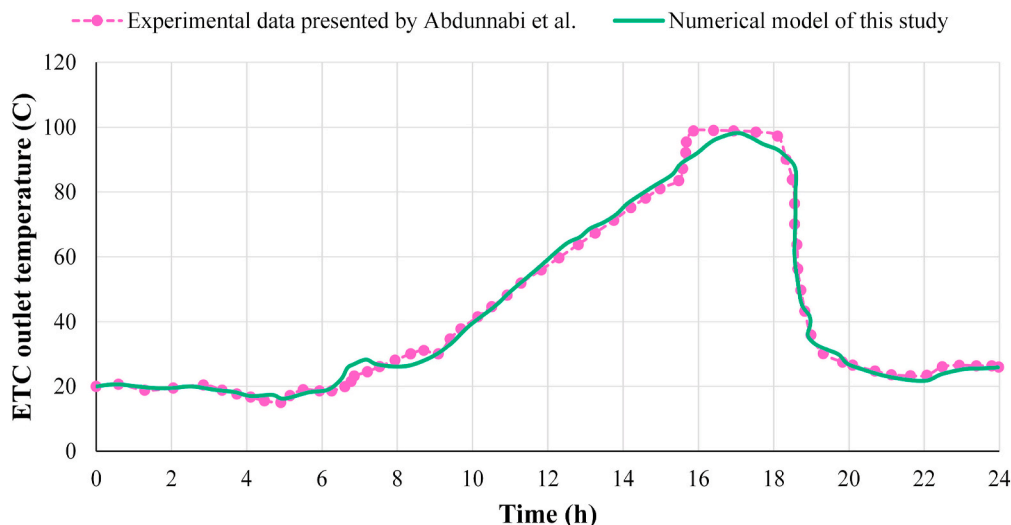


Fig. 5. Comparison of the numerical model of this study with experimental data presented by Abdunnabi et al. [51] for ETC outlet temperature.

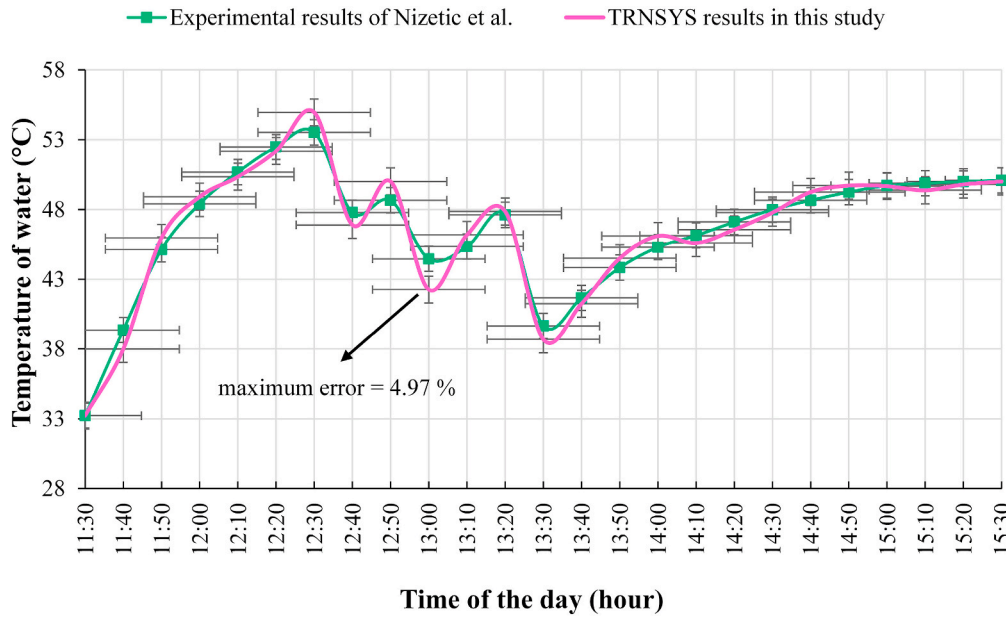


Fig. 6. Comparison of the TRNSYS simulation results in this study with experimental results presented by Nizetic et al. [53].

Table 5
Optimum results obtained by TRS-RSM-LCA strategy.

Optimum results		
Factors	Solar collector area	213.39 m ²
	Fuel cell capacity	28.35 kW
	Solar collector type	PVT
	Cooling system type	AC
Responses	NE	7581.64 kWh
	NN	40731.58 m ³
	LCC	396814.03 US\$
	GWP100	1396676.15 Eq. CO ₂
Reliability of proposed strategy	Composite desirability (CD) value	0.94
	Confirmation run (CR) maximum error	3.58%

its specific objectives by building on established knowledge, offering insightful contributions without redundantly verifying elements already deemed reliable.

3.2. Validation

Given the absence of a comparable experimental study to verify the numerical model developed for the hybrid system presented in this

Table 6
Comparing the optimum solutions and computational costs of the TRS-RSM-LCA optimization strategy and the genetic algorithm (GA) method.

Optimization factors	Results obtained from the TRS-RSM-LCA optimization strategy	Results obtained from the Genetic Algorithm (GA) optimization strategy	Difference between TRS-RSM-LCA results compared to GA results
solar collector area (m ²)	213.39	208.22	2.48 %
FC capacity (kW)	28.35	29.59	4.19 %
solar collector type/cooling system type	PVT/AC	PVT/AC	same
Computational time (s)	345	1908	81.9 % (reduction in computational costs compared to GA)

paper, the validation of the hybrid system's results relies on validating each individual component separately. Previous researchers have already validated the default components encoded in the TRNSYS and TESS libraries, also known as Types [20,22–24]. Furthermore, the reliability of these components has been established through their extensive utilization in prior studies [19]. Nevertheless, in order to ensure the accuracy of the TRNSYS and TESS library outcomes, the numerical method's results are validated in this study, specifically for the evacuated tube solar collector (ETC) and absorption chiller (AC) as exemplars.

Fig. 5 depicts the comparison of the numerical model of this study with experimental data presented by Abdunnabi et al. [51] for ETC outlet temperature. As shown, the numerical results of this study are in a good agreement with experimental results. Table 4 shows the comparison of the numerical model of this study with experimental data presented by He et al. [52] for system COP of AC and average room temperature provided by AC. As seen, the numerical results and experimental data are well-matched, with a maximum error of 6.7 % for the system COP on August 15th, and a maximum error of 5.9 % for the average room temperature on July 28th.

Fig. 6 presents a comparison between the TRNSYS simulation results from this study and the experimental results reported by Nizetic et al. [53]. The close alignment between the two data sets, with a maximum error of 4.97 %, demonstrates the accuracy of the simulation model in predicting water temperature throughout the day. Error bars indicate the variability in experimental measurements, further validating the reliability of the simulation approach.

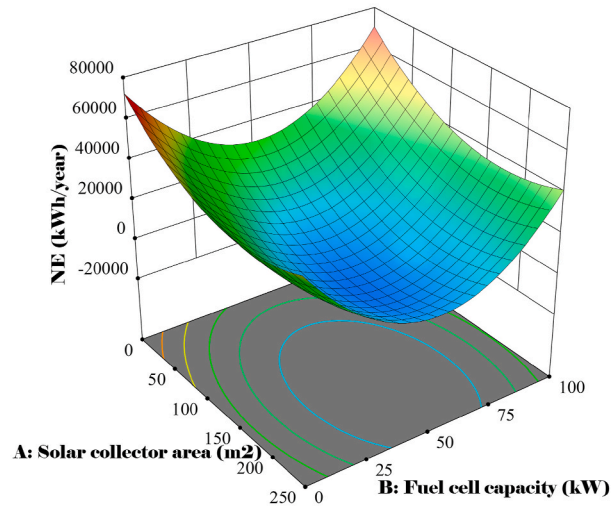
3.3. Optimization results

The traditional multi-objective optimization process typically involves first obtaining the Pareto optimal solutions, and then selecting a specific solution based on different decision-making strategies. However, in this study, a different approach employed using the TRS-RSM-LCA optimization strategy. This approach allows to efficiently identify the optimal system design without the need for a separate Pareto optimization stage. By leveraging the RSM and desirability function techniques, the optimal system configuration that collectively maximizes the 3E objectives can be directly obtained. As previously explained in section 3.1, The presented approach suggests conducting a total of 78 simulation cases to optimize the process, accounting for the selected factors in this study. The Design-Expert platform is utilized to perform

NE (kWh/year)
-9655.92 70000

X1 = A
X2 = B

Actual Factors
C = PVT
D = AC

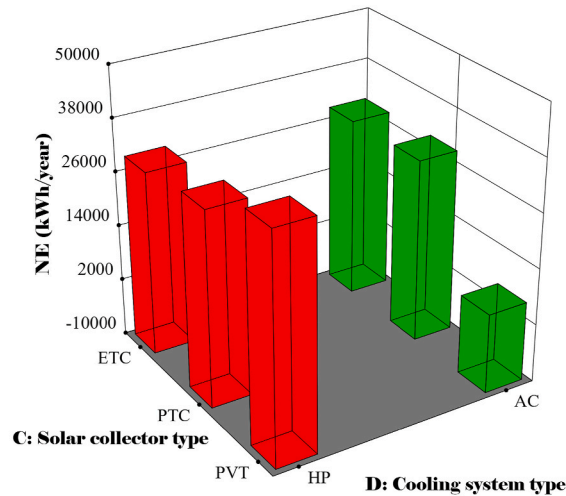


(a)

NE (kWh/year)

X1 = C
X2 = D

Actual Factors
A = 213.388
B = 28.3548



(b)

Fig. 7. The pattern of variations for the net electricity consumption (NE) in response to changes in two different sets of factors: (a) the solar collector area and fuel cell capacity, and (b) the solar collector type and cooling system type.

the necessary calculations using the response surface method and design of experiments (DOE) techniques.

The TRS-RSM-LCC optimization technique successfully determines the best system design by considering various factors. The optimized system, detailed in Table 5, incorporates factors that deliver superior performance based on the 3E criteria. Table 5 also presents the system's responses when using these optimized factors. The optimal system includes a solar collector area of 213.39 m², a fuel cell capacity of 28.35 kW, a PVT solar collector type, and an AC cooling system. It achieves the following metrics: NE of 7581.64 kWh.year⁻¹, NN of 40731.58 m³ year⁻¹, LCC of 396814.03 US\$, and GWP100 of 1396676.15 kg CO₂ eq.

The composite desirability (CD) value, which assesses the overall effectiveness of the optimization, is determined to be 0.94. This indicates that the optimal system is highly desirable across all factors

considered. Additionally, a confirmation run (CR) is conducted to validate the accuracy of the TRS-RSM-LCC technique by using the optimized parameters. The results show that the maximum error observed is 3.58 %, indicating that the proposed optimization technique achieves an acceptable level of accuracy with lower calculation costs. The confirmation run results, along with the CD metric, confirm the effectiveness and reliability of the employed optimization strategy.

In Table 6, a comparative analysis is provided between the proposed TRS-RSM-LCA optimization strategy and the Genetic Algorithm (GA) for the trigeneration system. The results show that both methodologies yield similar optimal solutions for key parameters, with only minimal differences observed—2.48 % for the solar collector area and 4.19 % for the fuel cell capacity—demonstrating the accuracy of the TRS-RSM-LCA approach. The system configuration is consistent across both methods,

recommending a PVT (Photovoltaic/Thermal) solar collector with an AC (absorption chiller) cooling system. A significant advantage of the TRS-RSM-LCA methodology is evident in its computational efficiency, achieving an 81.9 % reduction in computational time compared to GA (345 s versus 1908 s). This efficiency is due to the TRS-RSM-LCA strategy's use of Response Surface Methodology (RSM) for surrogate modeling, which minimizes the need for iterative simulations. However, certain limitations are noted, as the use of RSM and central composite design (CCD) may reduce flexibility in handling highly nonlinear or discrete variables. While the TRS-RSM-LCA is effective for typical optimization tasks, GA may offer advantages in complex solution landscapes by avoiding local minima more effectively. This comparative analysis highlights TRS-RSM-LCA's strengths in computational efficiency and accuracy, with an understanding of its limitations in specific

optimization scenarios.

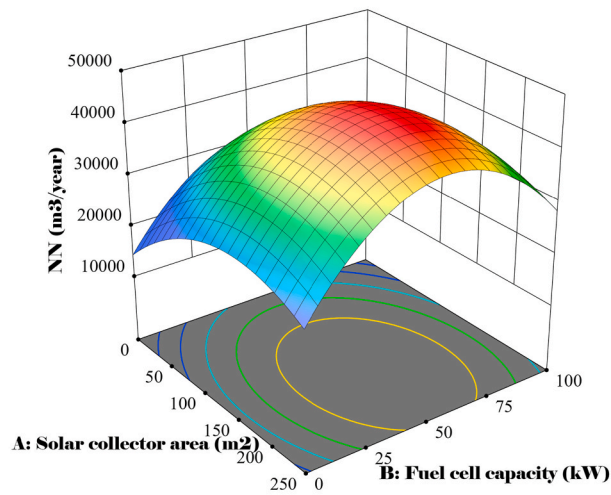
3.4. Effect of design factors on 3E responses

To evaluate the performance of the optimal system, a comparative analysis with baseline systems is important. However, selecting appropriate baseline systems can be challenging when dealing with a system that has many design variables. The number of potential baselines seems infinite, making it difficult to accurately assess the optimal system. To address this issue, this study incorporates a statistical approach into transient simulation and life cycle assessment. This integration enables the examination of interaction effects among design factors on the system's responses. By using statistical methods, valuable insights are gained regarding how different factors influence the overall

NN (m³/year)
24542.3 45469.2

X1 = A
X2 = B

Actual Factors
C = PVT
D = AC

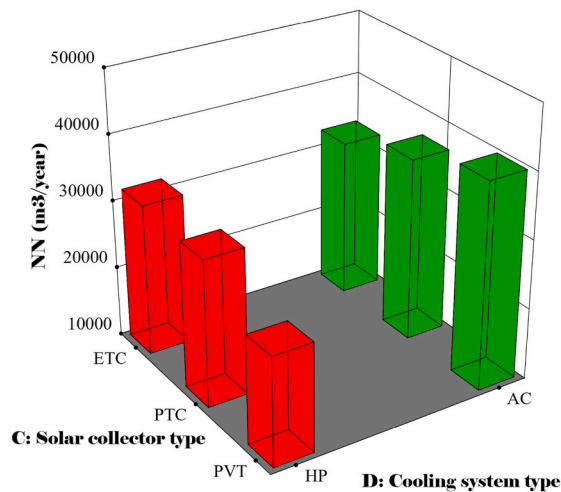


(a)

NN (m³/year)

X1 = C
X2 = D

Actual Factors
A = 213.388
B = 28.3548



(b)

Fig. 8. The pattern of variations for the net natural gas consumption (NN) in response to changes in two different sets of factors: (a) the solar collector area and fuel cell capacity, and (b) the solar collector type and cooling system type.

performance of the system. Fig. 7 through Fig. 10 illustrate the effects of selected factors interacting with each other on the 3E responses. Each figure focuses on two specific factors. This isolation allows for a better observation and analysis of the individual impacts of these factors on the system's responses.

The findings presented in Fig. 7 show the variations in net electricity consumption (NE) through a 3D response surface chart. The chart examines the impact of two sets of factors: (a) solar collector area and fuel cell capacity, and (b) solar collector type and cooling system type. Fig. 7a reveals that increasing the solar collector area from 0 to 250 m² leads to a significant reduction in NE, decreasing from approximately 70,000 kWh to around 25,000 kWh, representing a 64.3 % decrease. The optimal solar collector type is identified as PVT, and increasing the area of PVT leads to an increase in generated electricity and a decrease in NE.

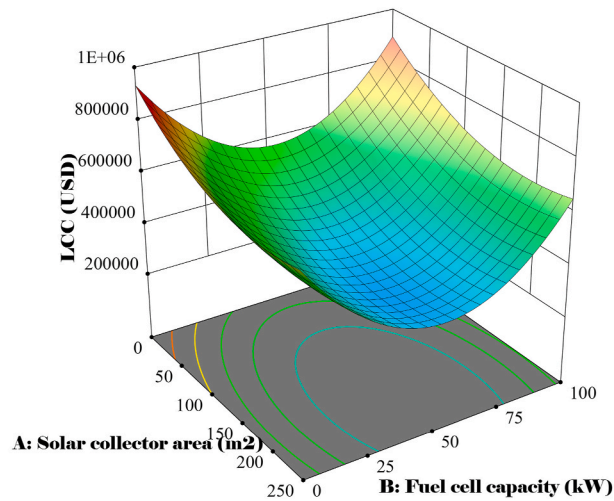
Examining changes in fuel cell capacity, Fig. 7a shows that increasing the capacity from 0 to 100 kW initially decreases NE. However, as the capacity continues to increase, NE starts to rise again, with the minimum point reached at approximately 40 kW. These findings emphasize the importance of optimizing both solar collector area and fuel cell capacity to reduce NE and improve efficiency in electricity generation. Choosing the optimal size of the solar collector and the appropriate capacity of the fuel cell is crucial for achieving favorable NE outcomes.

Fig. 7b depicts the effects of solar collector type and cooling system type on NE. The results show that using PVT collectors in combination with an absorption chiller (AC) leads to significantly lower NE compared to using a heat pump (HP) as the cooling system. This combination results in a remarkable reduction in NE of approximately 76.3 %. The use of PVT collectors, which combine photovoltaic and thermal

LCC (USD)
202513 900000

X1 = A
X2 = B

Actual Factors
C = PVT
D = AC

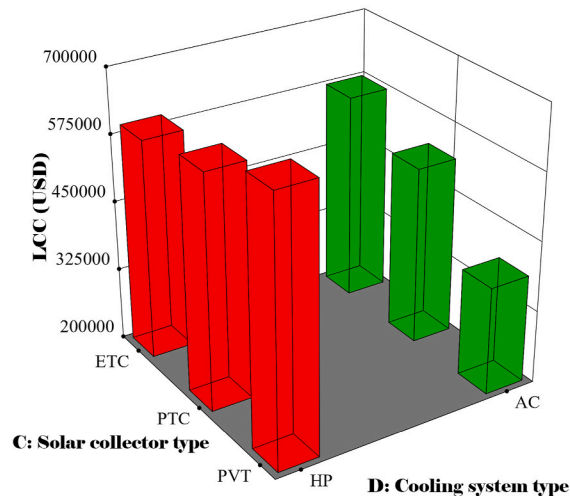


(a)

LCC (USD)

X1 = C
X2 = D

Actual Factors
A = 213.388
B = 28.3548



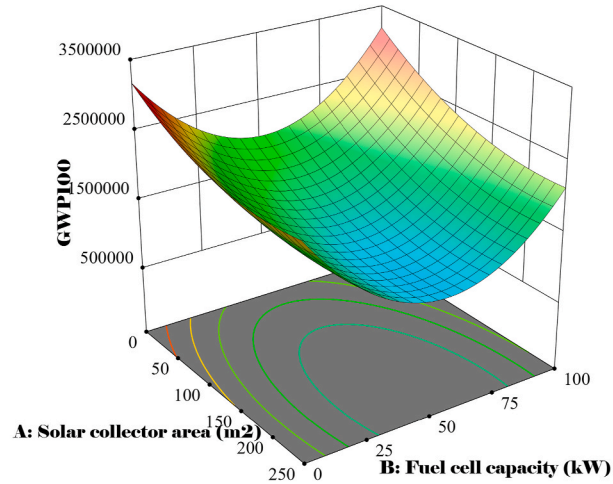
(b)

Fig. 9. The pattern of variations for the life cycle costs (LCC) in response to changes in two different sets of factors: (a) the solar collector area and fuel cell capacity, and (b) the solar collector type and cooling system type.

GWP100 (kg CO₂ eq.)
558587 3000000

X1 = A
X2 = B

Actual Factors
C = PVT
D = AC

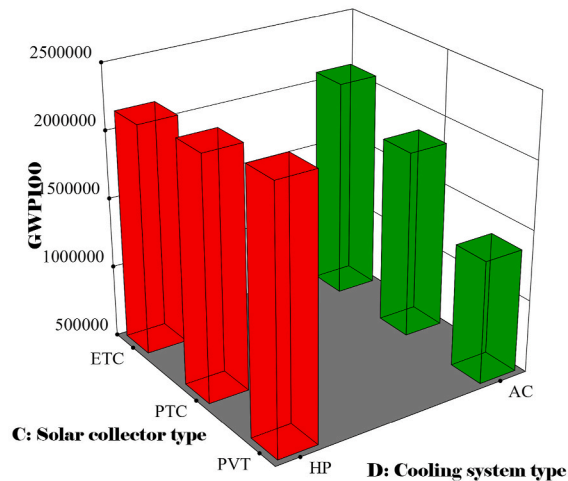


(a)

GWP100 (kg CO₂ eq.)

X1 = C
X2 = D

Actual Factors
A = 213.388
B = 28.3548



(b)

Fig. 10. The pattern of variations for the 100-year global warming potential (GWP100) in response to changes in two different sets of factors: (a) the solar collector area and fuel cell capacity, and (b) the solar collector type and cooling system type.

technologies, allows for efficient utilization of solar energy by generating both electricity and heat. The implementation of an absorption chiller, which utilizes waste heat or low-grade heat, further enhances energy efficiency by reducing the reliance on electricity for cooling purposes. In contrast, when a heat pump is used as the cooling system, NE is higher compared to the combination of PVT collectors and the absorption chiller. Heat pumps rely on electricity for transferring heat, which increases electricity consumption. These findings highlight the importance of selecting suitable combinations of solar collector type and cooling system type to optimize energy efficiency and reduce NE. They underscore the significance of considering the interplay between these factors in energy system design and decision-making processes.

Fig. 8 illustrates the NN variations in response to changes in two sets of factors. In Fig. 8a, it is observed that increasing the solar collector area leads to a decrease in NN, as more solar energy can be captured,

reducing the reliance on natural gas. However, when examining the impact of fuel cell capacity on NN, an interesting trend is observed. Initially, as fuel cell capacity increases from 0 to 100 kW, NN increases. However, beyond a certain threshold, NN starts to decrease, indicating an optimal fuel cell capacity that minimizes natural gas consumption. Importantly, the findings in Fig. 8 suggest that the best performance in terms of NN is achieved when both solar collector area and fuel cell capacity are maximized. By maximizing the solar collector area, more solar energy can be harnessed, reducing the dependence on natural gas. Simultaneously, optimizing the fuel cell capacity ensures efficient utilization of natural gas, further minimizing NN. These findings emphasize the importance of considering both solar collector area and fuel cell capacity as critical factors in reducing natural gas consumption.

In Fig. 8b, it is noteworthy that the combination of PVT collectors and AC leads to a higher NN. This increase is due to the additional

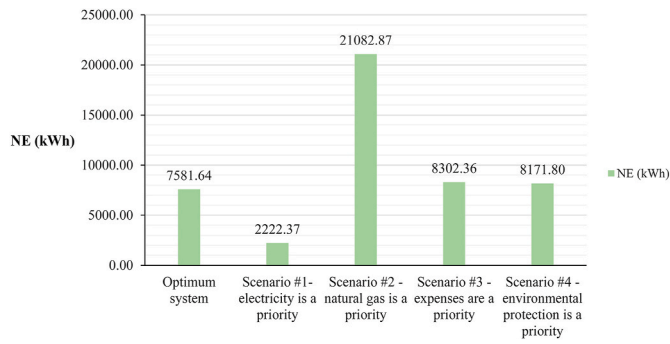


Fig. 11. Comparison of TEC for optimum systems under different weather conditions and base case scenarios.

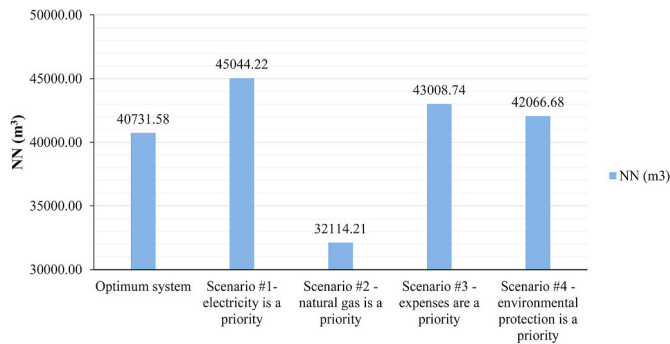


Fig. 12. Comparison of TNGC for optimum systems under different weather conditions and base case scenarios.

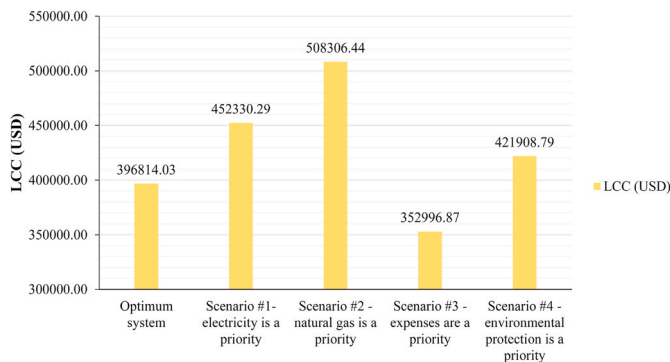


Fig. 13. Comparison of LCC for optimum systems under different weather conditions and base case scenarios.

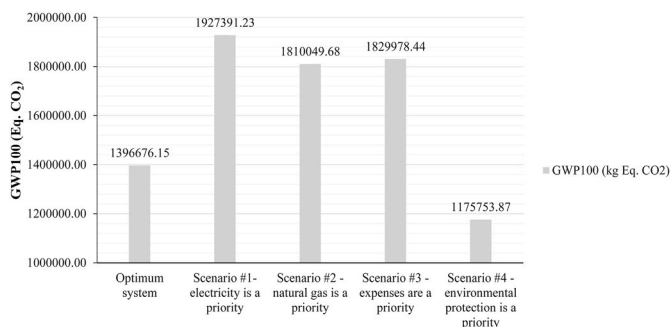


Fig. 14. Comparison of GWP100 for optimum systems under different weather conditions and base case scenarios.

natural gas required by the AC generator, which relies on the combustion of natural gas for boiler operation. However, the results also show that this high NN value can be mitigated by using evacuated tube collectors (ETC) instead. ETCs have higher thermal efficiency, which reduces the overall natural gas consumption needed for cooling when combined with the absorption chiller. By utilizing ETCs with the absorption chiller, a lower NN value can be achieved. Additionally, the results indicate that, in general, using HP instead of AC leads to lower NN. The results emphasize the importance of considering the interplay between solar collector type and cooling system type to achieve lower net natural gas consumption and promote more environmentally friendly energy systems.

Fig. 9 illustrates the variations in life cycle costs (LCC) in response to changes in two sets of factors: (a) solar collector area and fuel cell capacity, and (b) solar collector type and cooling system type. The figure shows that increasing the solar collector area from 0 to 250 results in a reduction in LCC by approximately 33.2 %. Additionally, increasing the fuel cell capacity from 0 to 100 kW achieves the minimum LCC value at around 45 kW. Furthermore, the results indicate that the combination of PVT solar collectors and AC significantly lowers the LCC compared to the combination of HP with ETC or PTC.

Fig. 10 shows the variations in 100-year global warming potential (GWP100) in response to changes in two factors: (a) solar collector area and fuel cell capacity, and (b) solar collector type and cooling system type. Increasing solar collector area reduces GWP100 by around 46.7 %, indicating the environmental benefits of relying more on clean, renewable solar energy. Optimizing fuel cell capacity, particularly at around 50 kW, minimizes the system's contribution to global warming. In terms of solar collector type and cooling system, the combination of PVT collectors and AC significantly lowers GWP100 compared to other combinations. This is due to efficient utilization of waste heat and clean energy generation. Overall, these findings highlight the importance of considering solar collector area, fuel cell capacity, and the combination of PVT collectors with AC in minimizing greenhouse gas emissions and addressing climate change. They provide valuable insights for designing cost-effective and environmentally sustainable energy systems.

3.5. Comparative analysis based on prioritizing scenarios

The optimum system is achieved by simultaneously optimizing factors such as electricity, natural gas, expenses, and environmental protections. To understand the trade-offs and synergies among these priorities, it is crucial to compare the optimal system with individual scenarios. This comparison helps identify potential conflicts and synergies and enables decision-makers to evaluate the overall performance of the system, considering interactions between electricity generation, natural gas usage, expenses, and environmental impacts. The following comparative analysis is conducted based on the 4 different prioritized scenarios: electricity as a priority, natural gas as a priority, expenses as a priority, and environmental protections as a priority (see Fig. 13).

Reducing net electricity consumption is the main goal in scenario #1, where electricity is a priority. This means determining the best mix of variables to meet the necessary energy demand while consuming the least amount of electricity. In scenario #1, prioritizing electricity consumption reduction yielded promising results. NE was significantly reduced by 70.7 % compared to the improved optimum system. However, there were trade-offs observed in other factors. NN increased by 10.6 %, indicating a potential conflict between electricity and natural gas priorities. LCC increased by 14.1 %, suggesting higher expenses associated with reducing electricity consumption. Moreover, the environmental impact, measured by GWP100, worsened by 38 % compared to the optimum system. Therefore, scenario #1 successfully reduces net electricity consumption, but there are trade-offs in other factors such as natural gas consumption, life cycle costs, and environmental impact. The specific values provided in Fig. 11 through Fig. 14 allow for a quantitative comparison between the scenario and the optimum system,

highlighting the areas where the scenario's priorities align or diverge from the overall optimized solution (see Fig. 12).

In scenario #2, the focus is on minimizing natural gas usage and associated costs. The objective is to reduce reliance on natural gas as much as possible. The analysis in Fig. 11 through Fig. 14 provides specific values for different factors compared to the optimum system. In scenario #2, the analysis reveals that prioritizing natural gas usage

worsens electricity consumption by 178.1 %. However, it successfully achieves a 21.2 % reduction in natural gas usage. There are trade-offs to consider, though. The scenario leads to a 28.1 % increase in costs (LCC) and a 29.6 % increase in environmental impact (GWP100) compared to the optimum system. In summary, scenario #2 achieves its goal of reducing natural gas consumption, but at the expense of increased electricity consumption, higher costs, and a greater environmental

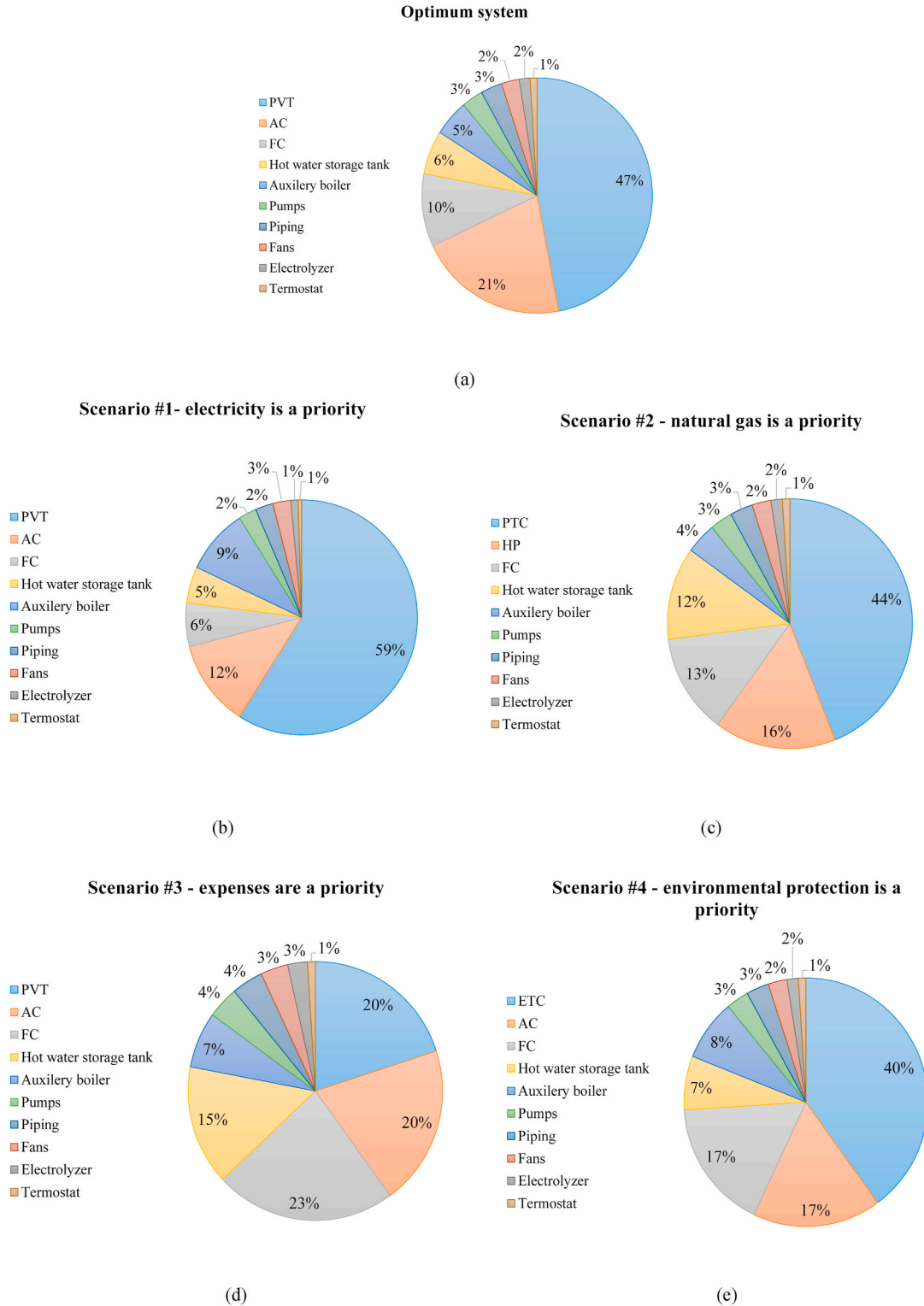


Fig. 15. The initial investment costs associated with different components of the hybrid system for: (a) optimum system, (b) Scenario #1, (b) Scenario #2, (b) Scenario #3, (b) Scenario #4.

impact.

In scenario #3, the priority is to minimize expenses and achieve strong economic performance. The goal is to reduce costs within the system. According to Fig. 11 through Fig. 14, in scenario #3, there is a slight increase of 9.5 % in electricity consumption (NE) compared to the

optimum system. Natural gas consumption (NN) increases by 5.6 %, which is still within an acceptable range considering the cost reduction focus. Impressively, the scenario achieves an 11 % reduction in costs (LCC) compared to the optimum system, indicating successful cost reduction and improved economic performance. However, there is a

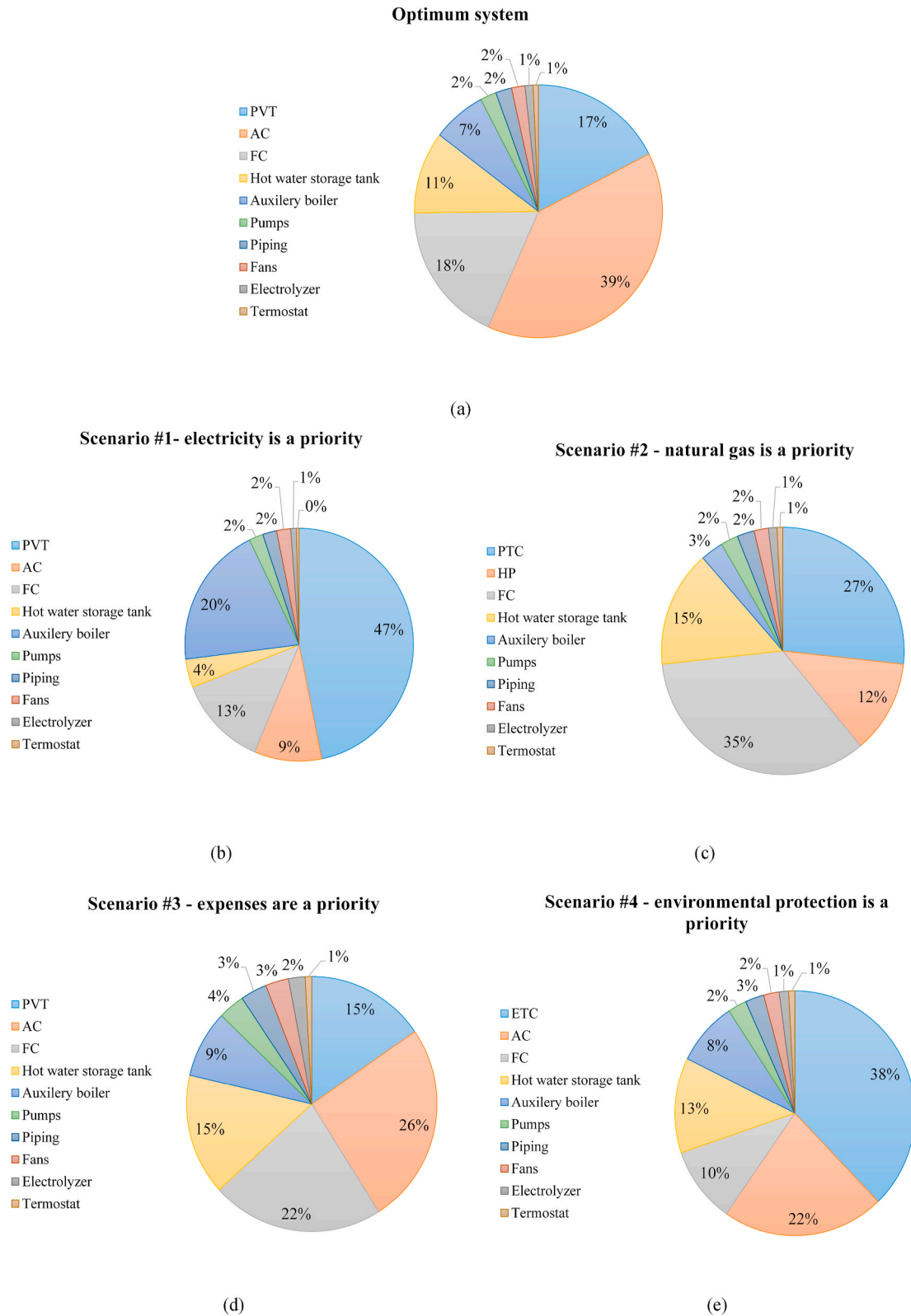


Fig. 16. The GWP100 associated with different components of the hybrid system for: (a) optimum system, (b) Scenario #1, (b) Scenario #2, (b) Scenario #3, (b) Scenario #4.

31.1 % increase in environmental impact (GWP100) compared to the optimum system, suggesting that prioritizing cost reduction may lead to a higher environmental impact.

Scenario #4 focuses on environmental protection and aims to minimize its impact on the environment. It prioritizes sustainability by optimizing energy consumption. In this scenario, the electricity consumption (NE) is slightly higher, by 7.8 %, compared to the optimum system. Natural consumption (NN) increases by 3.3 %. LCC is 6.3 % higher than the optimum system, indicating slightly higher costs. However, the scenario significantly reduces environmental impacts (GWP100) by 15.8 %, improving environmental sustainability. In summary, scenario #4 excels in prioritizing environmental protection by achieving minimal increases in electricity and natural gas consumption. Although it incurs slightly higher costs, it significantly reduces the environmental impact in terms of greenhouse gas emissions. Decision-makers can use the specific values in Fig. 11 through Fig. 14 to compare scenario #4 with the optimum system and evaluate its alignment with their sustainability goals.

The results of this study have significant implications for both policy and practice in optimizing combined cooling, heating, and power (CCHP) systems integrated with renewable energy. The findings demonstrate that prioritizing different objectives (e.g., electricity savings, natural gas reduction, cost-efficiency, or environmental sustainability) can lead to substantial trade-offs in performance metrics such as costs, environmental impact, and fuel consumption. For policymakers, these insights emphasize the importance of developing flexible, multi-criteria frameworks that allow stakeholders to balance energy efficiency, cost savings, and environmental goals. This multi-objective approach can aid in formulating policies that incentivize energy optimization while minimizing the negative impacts associated with prioritizing a single metric over others.

For practical applications, these results provide a roadmap for decision-makers in energy systems to tailor CCHP configurations based on specific operational goals. For example, in regions where electricity scarcity is a primary concern, prioritizing electricity reduction may be advisable despite potential increases in natural gas use and costs. Alternatively, in areas where environmental sustainability is prioritized, decision-makers may choose configurations that minimize greenhouse gas emissions, even at the expense of slightly higher costs or energy use. Overall, the study offers a framework that can be adapted to different regional or organizational priorities, allowing energy system designers and operators to make informed decisions that align with both policy goals and real-world constraints.

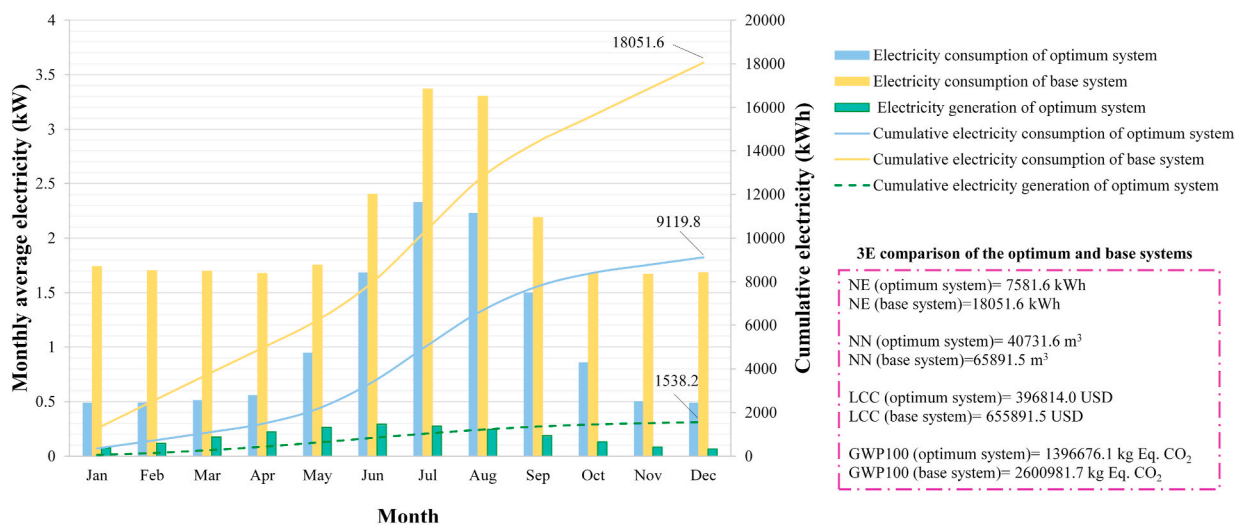


Fig. 17. the monthly average and cumulative electricity consumption for both the optimum and base systems, electricity generation in the optimum system as well as the 3E comparison of optimum and base systems.

Fig. 15 provides insights into the initial investment costs of the hybrid system components in the four prioritized scenarios. In the optimum system, PVT collectors account for 47 % of the investment, followed by AC at 21 % and the fuel cell (FC) at 10 %. Scenario #1, prioritizing electricity, has similar investment costs, with PVT collectors at 59 %, and AC and FC at 12 % and 6 % respectively. In scenario #2, prioritizing natural gas, there are significant changes, with PTC collectors at 44 % and HP at 16 %. Scenario #3, focusing on expenses, has the FC as the largest share at 23 %, and PVT collectors and AC both at 20 %. In scenario #4, prioritizing environmental protection, ETC has a share of around 40 %, while AC and FC each have 17 %. The comparative analysis shows the optimum system relies heavily on PVT collectors, while the prioritized scenarios vary based on their objectives, with changes in investment costs for different components.

Fig. 16 provides insights into the contributions of system components to the GWP100 in the selected scenarios. In the optimum system configuration AC contributes the most to the GWP100, accounting for 39 % of the total, followed by FC at 18 % and the PVT collectors at 17 %. However, when prioritizing electricity generation in scenario #1, the environmental impacts shift significantly. The PVT collectors become the predominant contributor, responsible for 47 % of the GWP100, while the auxiliary boiler and FC account for 20 % and 13 % respectively. In scenario #2, which prioritizes natural gas utilization, notable changes occur. The FC assumes a more prominent role, representing 35 % of the GWP100, while PTC contributes significantly at 27 %. Shifting the focus to expenses in scenario #3, the AC becomes the largest contributor with a share of 26 % in the GWP100. Additionally, both the auxiliary boiler and hot water storage tank contribute equally at 15 %. Lastly, scenario #4, which emphasizes environmental protection, reveals different shares among the components. ETC has the largest share, accounting for approximately 38 % of the GWP100, followed by the AC at 22 % and the hot water storage tank at 13 %. These results provide valuable insights into the varying environmental impacts and contributions of different components based on the prioritized scenarios.

3.6. Transient results

In the transient performance comparison between the optimum system and the base system, the base system represents a conventional approach where the cooling and heating demands of the building are met using an HP, and hot water demand is addressed with a boiler. The base system does not incorporate solar collectors or a hydrogen generation subsystem, unlike the optimized system.

Fig. 17 illustrates the monthly average and cumulative electricity consumption for both the optimum and base systems, as well as the electricity generation in the optimum system. In the optimum system, the cumulative electricity consumption over a year amount to 9119.8 kWh. This value is approximately 49.5 % lower than the electricity consumption of the base system, which stands at 18051.6 kWh. Besides, the optimum system has a yearly electricity generation of 1538.2 kWh (i. e., considering 9119.8 kWh consumption and 1538.2 kWh generation, net electricity consumption becomes 7581.6 kWh per year).

This significant reduction in electricity consumption highlights the superior efficiency and performance of the optimized system compared to the conventional base system. Overall, the transient performance comparison indicates that the optimized system significantly outperforms the base system in terms of electricity consumption, highlighting the advantages of incorporating solar collectors, absorption chiller and the hydrogen generation subsystem.

Fig. 17 also shows the 3E comparison for the optimum system and the base system. As shown in the bottom right of Fig. 15, the NE for the base system is 18,051.6 kWh, while this value for the optimum system is 7581.6 kWh. This shows around a 58 % net electricity reduction in the optimum system compared to the base system. The NN for the base system is 65,891.5 m³, while the NN for the optimum system is 40,731.6 m³, which is about 38.2 % lower compared to the base system. The LCC for the base system is 655,891.5 USD, while the LCC for the optimum system is 396,814.0 USD. This shows a 39.5 % reduction in costs compared to the base system. The GWP100 for the base system is 2,600,981.7 kg Eq. CO₂, while this value for the optimum system is 1,396,676.1 kg Eq. CO₂. This shows that the optimum system reduces environmental impacts by 46.3 % compared to the base system. Overall, Fig. 17 effectively illustrates the superior performance of the optimum system across various metrics, including net electricity consumption, net natural gas consumption, life cycle cost, and environmental impact. These results underscore the significant advantages and benefits of implementing the optimum system over the base system.

3.7. Future research direction

Future research could focus on exploring additional design parameters, such as varying types of renewable energy sources and storage options, to further enhance the flexibility and resilience of CCHP systems. Additionally, applying the TRS-RSM-LCA methodology to other hybrid systems or in different climatic regions could offer valuable insights into regional optimization strategies. Finally, incorporating advanced optimization techniques, such as machine learning algorithms, could enhance the accuracy and computational efficiency of the optimization process, providing a more comprehensive approach to sustainable energy system design.

4. Conclusions

In conclusion, this paper introduced the transient simulation-response surface method-life cycle assessment (TRS-RSM-LCA) as a groundbreaking optimization strategy for designing combined cooling, heat, and power (CCHP) systems that integrate solar collectors and hydrogen generation subsystems. This innovative approach not only targets the optimal energy, economic, and environmental (3E) performance of hybrid energy systems over their lifespan but also contributes meaningfully to the broader discourse on sustainable energy solutions.

By addressing critical global energy challenges—such as the need for increased energy efficiency and reduced greenhouse gas emissions—the findings from this study underscore the potential of CCHP systems to enhance energy security and sustainability. The optimization process considered both numerical factors, like solar collector area and fuel cell capacity, and categorical factors, including types of solar collectors and cooling systems. The TRS-RSM-LCA methodology successfully combined transient simulation, response surface method, and life cycle

assessment, creating a robust framework for optimizing hybrid energy systems.

The results revealed that the optimized CCHP system achieved a remarkable 49.5 % reduction in annual electricity consumption compared to the baseline configuration, demonstrating significant advancements in all 3E objectives. This study emphasized the importance of understanding interactions among design parameters and their effects on sustainability indicators. Additionally, the analysis of competing priorities offered valuable insights into potential synergies and conflicts, guiding stakeholders in comprehensive performance evaluations.

The optimal system configuration, identified through TRS-RSM-LCA, featured a solar collector area of 213.39 m², a fuel cell capacity of 28.35 kW, a photovoltaic/thermal (PVT) solar collector type, and an absorption chiller cooling system. This configuration produced favorable outcomes regarding net energy production, natural gas consumption, life cycle costs, and greenhouse gas emissions.

Overall, this research not only advances the design of CCHP systems by providing an effective optimization strategy but also serves as a vital resource for stakeholders committed to achieving efficient, sustainable hybrid energy systems that can address pressing global energy challenges.

CRedit authorship contribution statement

Mohammad Salahinezhad: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Hamed Tahami:** Writing – review & editing, Validation, Methodology. **Seyedeh Elham Kamali:** Writing – review & editing, Software, Conceptualization. **Ehsan Saedpanah:** Writing – original draft, Supervision, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] I. Karakurt, B.D. Avci, G. Aydin, Leveraging the trend analysis for modeling of the greenhouse gas emissions associated with coal combustion, *Environ. Sci. Pollut. Control Ser.* (2024) 1–25.
- [2] T. Ouyang, X. Tan, X. Tuo, P. Qin, C. Mo, Performance analysis and multi-objective optimization of a novel CCHP system integrated energy storage in large seagoing vessel, *Renew. Energy* 224 (2024) 120185.
- [3] Z. Ma, F. Dong, J. Wang, Y. Zhou, Y. Feng, Optimal design of a novel hybrid renewable energy CCHP system considering long and short-term benefits, *Renew. Energy* 206 (2023) 72–85.
- [4] X. Zhu, X. Zhan, H. Liang, X. Zheng, Y. Qiu, J. Lin, et al., The optimal design and operation strategy of renewable energy-CCHP coupled system applied in five building objects, *Renew. Energy* 146 (2020) 2700–2715.
- [5] P. Nejat, F. Jomehzadeh, M.M. Taheri, M. Gohari, M.Z.A. Majid, A global review of energy consumption, CO₂ emissions and policy in the residential sector (with an overview of the top ten CO₂ emitting countries), *Renewable and sustainable energy reviews* 43 (2015) 843–862.
- [6] N. Zheng, H. Zhang, L. Duan, X. Wang, Q. Wang, L. Liu, Multi-criteria performance analysis and optimization of a solar-driven CCHP system based on PEMWE, SOFC, TES, and novel PVT for hotel and office buildings, *Renew. Energy* 206 (2023) 1249–1264.
- [7] Y. Xiao, C. Zou, M. Dong, H. Chi, Y. Yan, S. Jiang, Feasibility study: economic and technical analysis of optimal configuration and operation of a hybrid CSP/PV/wind power cogeneration system with energy storage, *Renew. Energy* 225 (2024) 120273.
- [8] A.M. García, J. Gallagher, J.A.R. Díaz, A. McNabola, An economic and environmental optimization model for sizing a hybrid renewable energy and battery storage system in off-grid farms, *Renew. Energy* 220 (2024) 119588.
- [9] F. Ren, Z. Wei, X. Zhai, Multi-objective optimization and evaluation of hybrid CCHP systems for different building types, *Energy* 215 (2021) 119096.
- [10] S. Ghorbani, M. Deymi-Dashtebayaz, D. Dadpour, M. Delpisheh, Parametric study and optimization of a novel geothermal-driven combined cooling, heating, and power (CCHP) system, *Energy* 263 (2023) 126143.
- [11] J. Wang, Y. Lu, Y. Yang, T. Mao, Thermodynamic performance analysis and optimization of a solar-assisted combined cooling, heating and power system, *Energy* 115 (2016) 49–59.

- [12] Z. Song, T. Liu, Q. Lin, Multi-objective optimization of a solar hybrid CCHP system based on different operation modes, *Energy* 206 (2020) 118125.
- [13] N. Poyyamozi, S.S. Kumar, R.A. Kumar, G. Soundararajan, An investigation into enhancing energy storage capacity of solar ponds integrated with nanoparticles through PCM coupling and RSM optimization, *Renew. Energy* 221 (2024) 119733.
- [14] N. Mahdavi, P. Mojaver, S. Khalilarya, Multi-objective optimization of power, CO₂ emission and exergy efficiency of a novel solar-assisted CCHP system using RSM and TOPSIS coupled method, *Renew. Energy* 185 (2022) 506–524.
- [15] E. Saedpanah, M. Lahonian, M.Z.M. Abad, Optimization of multi-source renewable energy air conditioning systems using a combination of transient simulation, response surface method, and 3E lifespan analysis, *Energy* (2023) 127200.
- [16] N. Mao, B. Zhang, M. Song, S. Deng, A simplified numerical study on the energy performance and thermal environment of a bedroom TAC system, *Energy Build.* 166 (2018) 305–316.
- [17] F. Tooryan, H. HassanzadehFard, V. Dargahi, S. Jin, A cost-effective approach for optimal energy management of a hybrid CCHP microgrid with different hydrogen production considering load growth analysis, *Int. J. Hydrogen Energy* 47 (2022) 6569–6585.
- [18] M. Nedaei, M. Faccio, M. Bortolini, M. Gamberi, Design and simulation-based optimisation of a renewable energy system consisting of wind and hydrogen for the purpose of reliable electrification, *Int. J. Environ. Technol. Manag.* 25 (2022) 445–465.
- [19] R. Shrivastava, V. Kumar, S. Untawale, Modeling and simulation of solar water heater: a TRNSYS perspective, *Renewable and sustainable energy reviews* 67 (2017) 126–143.
- [20] G. A'lvarez, M. Chagolla, J. Xama' n, M. Jime' nez, S. Sua' rez, M. Heras, A TRNSYS simulation and experimental comparison of the thermal behavior of a building located in desert climate, *Energy Sustainability* (2010) 349–356.
- [21] J.F. Kreider, F. Kreith, *Solar Energy Handbook*, 1981.
- [22] W. Ya'ici, E. Entchev, K. Lombardi, Experimental and simulation study on a solar domestic hot water system with flat-plate collectors for the Canadian climatic conditions, *Energy Sustainability. American Society of Mechanical Engineers* (2012) 69–78.
- [23] Y. Allard, M. Kummert, M. Bernier, A. Moreau, Intermodel comparison and experimental validation of electrical water heater models in TRNSYS. *Proceedings of Building Simulation*, 2011, pp. 688–695.
- [24] D. Brough, J. Ramos, B. Delpach, H. Jouhara, Development and validation of a TRNSYS type to simulate heat pipe heat exchangers in transient applications of waste heat recovery, *International Journal of Thermofluids* 9 (2021) 100056.
- [25] M. Rezvanpour, D. Boroghani, F. Torabi, M. Pazoki, Using CaCl₂·6H₂O as a phase change material for thermo-regulation and enhancing photovoltaic panels' conversion efficiency: experimental study and TRNSYS validation, *Renew. Energy* 146 (2020) 1907–1921.
- [26] S.A. Kalogirou, R. Agathokleous, G. Barone, A. Buonomano, C. Forzano, A. Palombo, Development and validation of a new TRNSYS Type for thermosiphon flat-plate solar thermal collectors: energy and economic optimization for hot water production in different climates, *Renew. Energy* 136 (2019) 632–644.
- [27] S. Klein, B. Newton, J. Thornton, D. Bradley, J. Mitchell, M. Kummert, *TRNSYS Reference Manual: Mathematical Reference*, 2006.
- [28] TESS. Component Libraries v. 17, 01 for TRNSYS V17. 0 and the TRNSYS Simulation Studio, Parameter/Input/Output Reference Manual, Thermal Energy System Specialists, LLC, 2004.
- [29] E. Assareh, A. Dejdard, A. Ershadi, M. Jafarian, M. Mansouri, E. Azish, et al., Performance analysis of solar-assisted-geothermal combined cooling, heating, and power (CCHP) systems incorporated with a hydrogen generation subsystem, *J. Build. Eng.* 65 (2023) 105727.
- [30] E. Assareh, A. Dejdard, A. Ershadi, M. Jafarian, M. Mansouri, E. Azish, et al., Techno-economic analysis of combined cooling, heating, and power (CCHP) system integrated with multiple renewable energy sources and energy storage units, *Energy Build.* (2022) 112618.
- [31] N. Sommerfeldt, P. Ollas, Reverse engineering prototype solar PV/thermal collector properties from empirical data for use in TRNSYS type 560. *ISES Solar World Congress and IEA Solar Heating and Cooling Conference 2017*, 2017, pp. 1121–1132. Abu Dhabi, UAE, October 29–November 2, 2017.
- [32] I. González-Pino, E. Pérez-Iribarren, A. Campos-Celador, J. Terés-Zubiaga, Analysis of the integration of micro-cogeneration units in space heating and domestic hot water plants, *Energy* 200 (2020) 117584.
- [33] F. Oueslati, Hybrid renewable system based on solar wind and fuel cell energies coupled with diesel engines for Tunisian climate: TRNSYS simulation and economic assessment, *Int. J. Green Energy* 18 (2021) 402–423.
- [34] E. Saedpanah, H. Pasharsahri, Performance assessment of hybrid desiccant air conditioning systems: a dynamic approach towards achieving optimum 3E solution across the lifespan, *Energy* 234 (2021) 121151.
- [35] M. Mahmoudian, F. Goharpey, M. Behzadnasab, Z. Daneshfar, Shear thickening and hysteresis in dense suspensions: the effect of particle shape, *Journal of Rheology* 68 (2024) 479–490.
- [36] Y. Xie, P. Hu, N. Zhu, F. Lei, L. Xing, L. Xu, Collaborative optimization of ground source heat pump-radiant ceiling air conditioning system based on response surface method and NSGA-II, *Renew. Energy* 147 (2020) 249–264.
- [37] B.S. Qader, E. Supeni, M. Ariffin, A.A. Talib, RSM approach for modeling and optimization of designing parameters for inclined fins of solar air heater, *Renew. Energy* 136 (2019) 48–68.
- [38] H. Zamani, M. Moghiman, A. Kianifar, Optimization of the parabolic mirror position in a solar cooker using the response surface method (RSM), *Renew. Energy* 81 (2015) 753–759.
- [39] D.C. Montgomery, *Design and Analysis of Experiments*, John Wiley & sons, 2017.
- [40] G. Derringer, R. Suich, Simultaneous optimization of several response variables, *J. Qual. Technol.* 12 (1980) 214–219.
- [41] Design Expert Software, V 13.0.5, Stat-Ease Inc, Minneapolis (USA), 2021.
- [42] B.K. Sharma, M.K. Chandel, Life cycle cost analysis of municipal solid waste management scenarios for Mumbai, India, *Waste Management* 124 (2021) 293–302.
- [43] R.M. Nonki, E. Amoussou, C.J. Lennard, A. Lenou, R.M. Tshimanga, C. Houndenou, Quantification and allocation of uncertainties of climate change impacts on hydropower potential under 1.5° C and 2.0° C global warming levels in the headwaters of the Benue River Basin, Cameroon, *Renew. Energy* 215 (2023) 118979.
- [44] H. Aygun, H. Caliskan, Environmental and enviroeconomic analyses of two different turbofan engine families considering landing and take-off (LTO) cycle and global warming potential (GWP) approach, *Energy Convers. Manag.* 248 (2021) 114797.
- [45] L. Qiu, L. He, Y. Kang, D. Liang, Assessment of the potential of enhanced geothermal systems in Asia under the impact of global warming, *Renew. Energy* 194 (2022) 636–646.
- [46] H.H. Balyani, A. Sohani, H. Sayyaadi, R. Karami, Acquiring the best cooling strategy based on thermal comfort and 3E analyses for small scale residential buildings at diverse climatic conditions, *Int. J. Refrig.* 57 (2015) 112–137.
- [47] A. Sohani, H. Sayyaadi, Design and retrofit optimization of the cellulose evaporative cooling pad systems at diverse climatic conditions, *Appl. Therm. Eng.* 123 (2017) 1396–1418.
- [48] A. Sohani, H. Sayyaadi, N. Mohammadhosseini, Comparative study of the conventional types of heat and mass exchangers to achieve the best design of dew point evaporative coolers at diverse climatic conditions, *Energy Convers. Manag.* 158 (2018) 327–345.
- [49] M. Salimi, D. Faramarzi, S.H. Hosseini, G.B. Gharehpetian, Replacement of natural gas with electricity to improve seismic service resilience: an application to domestic energy utilities in Iran, *Energy* 200 (2020) 117509.
- [50] A. Sohani, H. Sayyaadi, M. Azimi, Employing static and dynamic optimization approaches on a desiccant-enhanced indirect evaporative cooling system, *Energy Convers. Manag.* 199 (2019) 112017.
- [51] M. Abdunnabi, K. Alakder, N. Alkishriwi, S. Abughres, Experimental validation of forced circulation of solar water heating systems in TRNSYS, *Energy Proc.* 57 (2014) 2477–2486.
- [52] T. He, X. Zhang, C. Wang, M. Wang, B. Li, N. Xue, et al., Application of solar thermal cooling system driven by low temperature heat source in China, *Energy Proc.* 70 (2015) 454–461.
- [53] S. Nizetic, D. Coko, I. Marasovic, Experimental study on a hybrid energy system with small-and medium-scale applications for mild climates, *Energy* 75 (2014) 379–389.